



UNIVERSITÉ GRENOBLE ALPES / ENSIMAG

Project Study Report

Wavelet Analysis and Synthesis of Fractional Brownian Motion

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Abstract

In this project, we explore several approaches for analyzing and synthesizing fractional Brownian motion (fBm). After recalling the theoretical background necessary to understand the subject, we focus on the statistical properties of fractional Brownian motion, with particular emphasis on the behavior of its wavelet coefficients and the estimation of the Hurst parameter using the Haar basis. We then investigate different synthesis strategies for fBm, based on the three methods introduced by *P.Flandrin* in [\[1\]](#) (our reference article). These approaches are analyzed and compared in terms of their ability to reproduce the defining properties of fractional Brownian motion.

Summary of the article

The objective of the article “*Wavelet Analysis and Synthesis of Fractional Brownian Motion*” [1] is to show how fractional Brownian motion (fBm) can be analyzed through wavelets; that is, how its main properties — self-similarity, long-term dependence, and its $1/f$ spectrum — are reflected in the statistical behavior of wavelet coefficients. More precisely, the $1/f$ spectral property of fBm manifests in the variance of wavelet coefficients across scales, which follows a power law of the form a^{2H+1} , where H is the Hurst exponent controlling the roughness or smoothness of the trajectory.

The paper also proposes a method to synthesize fBm, meaning to generate sample paths that are statistically equivalent to fBm using its wavelet representation. Furthermore, the author studies in depth the statistical properties of fBm’s wavelet coefficients, including their variance, correlations across scales, and inter-scale dependencies. Thanks to the multiscale structure of wavelets and their vanishing moments, these coefficients are almost non-correlated, which simplifies the analysis of long-range dependencies.

Overall, the article demonstrates both the analytical advantages of the wavelet representation for understanding fBm and a practical approach for reconstructing processes that preserve its key statistical properties.

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1 A Brief History of Fractional Brownian Motion

First studied in connection with self-similar stochastic processes by the Russian probabilist *Andrey Nikolaevich Kolmogorov* in the 1940s, *fractional Brownian motion* (fBm) is a generalization of Brownian motion that introduces an additional parameter—the Hurst exponent. This parameter controls both the degree of self-similarity and the long-range dependence of the process, thereby extending the scaling and correlation properties of classical Brownian motion.

Later, in 1968, the well-known Russian mathematician *Benoît Mandelbrot*, who discovered curious objects called *fractals*, wrote a paper [5] with *John W. Van Ness* on their study of fractional Brownian motion, aiming to provide a proper definition together with its main properties, and to supply a framework for modeling time series. They believed that such a process would be useful in many fields where these time series play a central role, such as finance, for instance.

Now, as is often the case in probability and statistics, the first question that naturally arises is whether the defined random object can actually be generated or simulated, and if so, by what means. It turns out that there are two main approaches for synthesizing fBm. The first one is the *Cholesky decomposition method* which relies on the inverse of the covariance matrix of the increments. However, it becomes computationally expensive, scaling cubically with the size of the matrix, as soon as the Brownian motion is no longer standard, since the covariance matrix is no longer diagonal. This is due, as we will see later, to the fact that the increments of fBm are only independent in the case of Brownian motion, which makes this method somewhat inconvenient for fBm in general.

Thus, this motivated the development of more sophisticated methods, such as the *approximate circulant method*, which is approximate by design, or, more accurately, the exact *circulant embedding method* introduced by Dietrich & Newsam in 1997.

2 Construction of Fractional Brownian Motion

2.1 A Short Overview of Self-Similarity

The notion of self-similarity emerged when people attempted to properly define *fractals* as mathematical objects, although even today there is no universally agreed-upon definition. What can be said, however, is that fractals possess the property of self-similarity, meaning that they are locally composed of "smaller versions of themselves." As a result, they exhibit details at every scale, since these details are, in some sense, miniature versions of the whole structure.

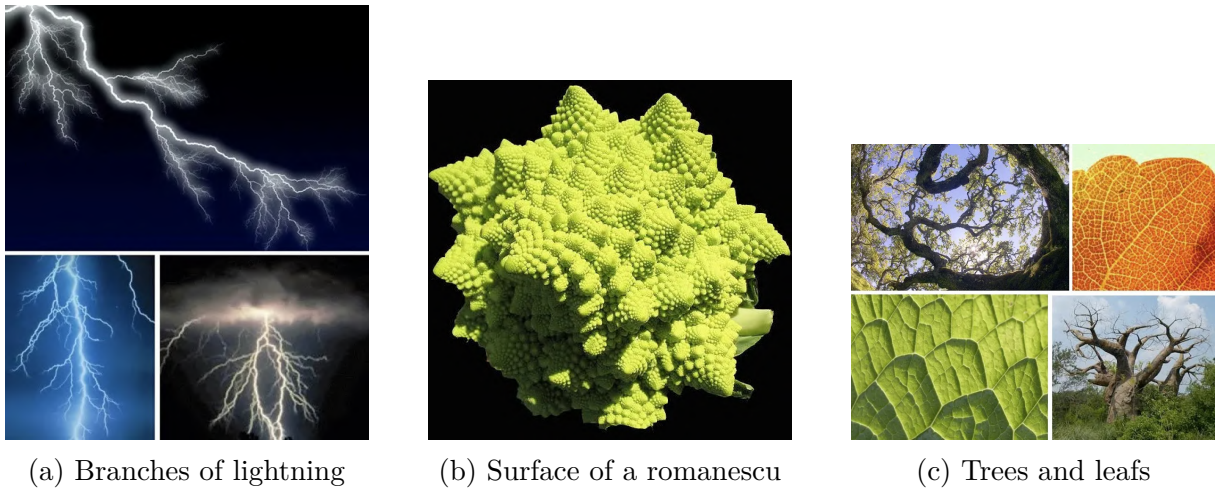


Figure 1: Three different examples of fractals found in nature. Source : [WebEcoist, 2008](#).

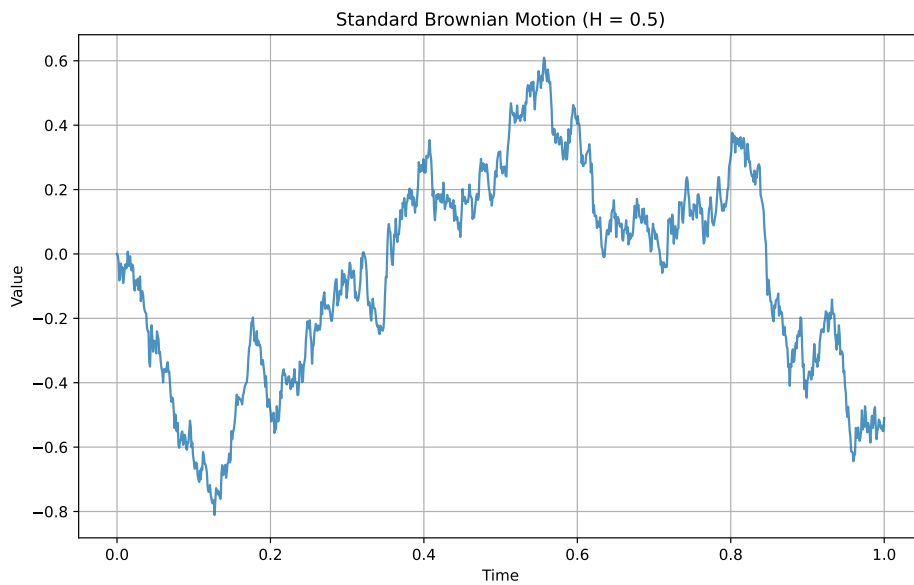


Figure 2: Realization of a fractional Brownian motion

As seen in Figure 1, fractals can be observed in everyday life, although their structures do not repeat endlessly as described in theory. However, this does not really matter, since beyond a certain point further details are no longer visible as the changes become too small, so that this can be regarded as a satisfying approximation. Moreover, self-similarity can also be seen in curves and graphs, especially those arising from random processes and time series. Fractional Brownian motion is a typical example, as seen in figure 2, and happens to be the focus of this study.

2.2 Stochastic processes in a nutshell

First, we shall introduce some theory regarding stochastic processes that leads to the characterization of fractional Brownian motion.

Definition 2.1 (Stochastic process).

Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space, and $(E, \mathcal{B})$ a measurable space.

A *stochastic process* indexed by T and valued in E is a family of random variables $(X_t)_{t \in T}$ such that :

$$\forall t \in T, \quad X_t : \omega \in \Omega \mapsto X_t(\omega) \in E \quad \text{is measurable}$$

If T is an interval of $\mathbb{R}$ (e.g., $T = [0, \infty)$), the process is called a *continuous-time process*.

Definition 2.2 (Sample path / Trajectory).

A *sample path* (or *trajectory*) of a stochastic process $(X_t)_{t \in T}$ is the function :

$$t \in T \mapsto X_t(\omega) \in E$$

for a fixed outcome $\omega \in \Omega$. It represents the realization of the process for one experiment.

Definition 2.3 (Gaussian stochastic process).

A stochastic process $(X_t)_{t \in T}$ is called *Gaussian* if, for any finite set of times $t_1, \dots, t_n \in T$, the random vector $(X_{t_1}, \dots, X_{t_n})$ follows a multivariate normal distribution. That is, it is completely determined by its mean vector m and covariance matrix Σ :

$$m := (\mathbb{E}[X_{t_i}])_{1 \leq i \leq n}, \quad \Sigma := (\Sigma_{i,j} := \text{Cov}(X_{t_i}, X_{t_j}))_{1 \leq i,j \leq n}$$

Definition 2.4 (Self-similar stochastic processes).

A stochastic process $(X_t)_{t \in \mathbb{R}}$ is said to be *H-self-similar* with *Hurst exponent* $H > 0$ if :

$$\forall \varepsilon > 0, \quad (X_{\varepsilon t})_{t \in \mathbb{R}} \stackrel{(d)}{=} (\varepsilon^H X_t)_{t \in \mathbb{R}},$$

where $\stackrel{(d)}{=}$ denotes equality in distribution. This means the statistical properties are invariant under suitable scaling of time and amplitude.

Definition 2.5 (Stationarity).

Let $(X_t)_{t \in \mathbb{R}}$ be a stochastic process.

1. **Stationarity** : A stochastic process $(X_t)_{t \in \mathbb{R}}$ is said to be *stationary* if its statistical properties do not change over time, meaning that :

$$\forall s, t \in \mathbb{R}, \quad X_s \stackrel{(d)}{=} X_t$$

2. **Strict stationarity** : The process is *strictly stationary* if, for all $n \geq 1$, and for all $h \geq 0$:

$$\forall t_1, t_2, \dots, t_n \in \mathbb{R}, \quad (X_{t_1+h}, \dots, X_{t_n+h}) \stackrel{(d)}{=} (X_{t_1}, \dots, X_{t_n}).$$

3. **Stationary increments** : The process has *stationary increments* if, for all $h \geq 0$:

$$\forall t \in \mathbb{R}, \quad X_{t+h} - X_t \stackrel{(d)}{=} X_h - X_0.$$

Theorem 2.6 (Characterization of Self-Similar Processes with Stationary Increments).

Let $(X_t)_{t \in \mathbb{R}}$ be a non-constant real-valued H -self-similar stochastic process with stationary increments. If one assumes that :

1. $(X_t)_{t \in \mathbb{R}}$ is a second-order stochastic process, i.e :

$$\forall t \in \mathbb{R}, \quad \mathbb{E}[X_t^2] < +\infty$$

2. $(X_t)_{t \in \mathbb{R}}$ is continuous in distribution at time $t = 0$, i.e :

$$X_\varepsilon \xrightarrow[\varepsilon \rightarrow 0]{(d)} X_0$$

Then :

1. $0 \leq H \leq 1$, and if $H > 0$ then $X_0 \stackrel{(d)}{=} 0$ almost surely, i.e. :

$$\text{For almost all } \omega \in \Omega, \quad X_0(\omega) = 0$$

2. If $0 < H < 1$, $(X_t)_{t \in \mathbb{R}}$ is a centered process, i.e. :

$$\forall t \in \mathbb{R}, \quad \mathbb{E}[X_t] = 0$$

3. For $H = 1$:

$$\forall t \in \mathbb{R}, \quad X_t = tX_1 \quad \text{almost surely}$$

4. its covariance is given, up to a multiplicative constant, by :

$$\forall s, t \in \mathbb{R}, \quad \text{Cov}(X_s, X_t) = \begin{cases} \frac{\mathbb{E}[X_1^2]}{2} (|s|^{2H} + |t|^{2H} - |s - t|^{2H}) & \text{if } 0 < H < 1 \\ \text{Var}(X_1)st & \text{if } H = 1 \end{cases}$$

Proof. 1. Let $\varepsilon > 0$. Then, for $H \neq 0$, the process $(X_t)_{t \in \mathbb{R}}$ being H -self-similar :

$$X_0 \stackrel{(d)}{=} \varepsilon^H X_0 \underbrace{\Longleftrightarrow}_{H \neq 0} X_0 \stackrel{(d)}{=} 0 \iff \text{For almost all } \omega \in \Omega, \quad X_0(\omega) = 0$$

Now, let us prove that $0 \leq H \leq 1$ by considering a sequence $(\varepsilon_n)_{n \geq 0} \in (\mathbb{R}_+^*)^{\mathbb{N}}$ which converges to 0.

Therefore, since :

$$\forall n \geq 0, \quad X_{\varepsilon_n} \stackrel{(d)}{=} \varepsilon_n^H X_1$$

and as $X_{\varepsilon_n} \xrightarrow[n \rightarrow +\infty]{(d)} X_0 \stackrel{(d)}{=} 0$, $(\varepsilon_n^H)_{n \geq 0}$ is required to converge to 0 as well, which thus implies that $H \geq 0$. To prove that $H \leq 1$, one first notices that :

$$\forall t \in \mathbb{R}, \quad X_{t+1} - X_t \stackrel{(d)}{=} X_1 - X_0 \stackrel{(d)}{=} X_1$$

which yields the following :

$$\mathbb{E}[|X_2|] = \mathbb{E}[|X_2 - X_1 + X_1|] \leq \mathbb{E}[|X_2 - X_1|] + \mathbb{E}[|X_1|] = 2\mathbb{E}[|X_1|]$$

Hence, using the H -self-similarity of $(X_t)_{t \in \mathbb{R}}$, one obtains that :

$$2^H \mathbb{E}[|X_1|] \leq 2\mathbb{E}[|X_1|]$$

Consequently, if $\mathbb{E}[|X_1|] \neq 0$, then $2^{H-1} \leq 1$ which further implies that $H \leq 1$. However, by assuming that $H > 0$, since the process has stationary increments, we have :

$$\begin{aligned} X_1 &\stackrel{(d)}{=} X_0 - X_{-1} \stackrel{(d)}{=} -X_{-1} \\ \implies \forall t < 0, \quad X_t &\stackrel{(d)}{=} (-t)^H X_{-1} \stackrel{(d)}{=} -(-t)^H X_1 \\ \implies \forall t \in \mathbb{R}^*, \quad X_t &\stackrel{(d)}{=} |t|^{H-1} t X_1 \end{aligned} \tag{2.2.1}$$

If $\mathbb{E}[|X_1|] = 0$, then $X_1 = 0$ almost surely. By (2.2.1) and since $X_0 \stackrel{(d)}{=} 0$, this implies that the process is identically zero, and therefore $\mathbb{E}[|X_1|] \neq 0$.

2. Let $0 < H < 1$. Then, since :

$$\mathbb{E}[X_1] = \mathbb{E}[X_1 - X_0] = \mathbb{E}[X_2 - X_1] = \underbrace{(2^H - 1)}_{\neq 1} \mathbb{E}[X_1] \iff \mathbb{E}[X_1] = 0$$

one concludes from (2.2.1) that $(X_t)_{t \in \mathbb{R}}$ is centered.

3. By taking $H = 1$, one has that :

$$\forall t \in \mathbb{R}, \quad \|X_t - tX_1\|_{L^2_{\mathbb{P}}}^2 = \mathbb{E} [X_t^2 - 2tX_tX_1 + t^2X_1^2] = \mathbb{E}[t^2X_1^2 - 2t^2X_1^2 - t^2X_1^2] = 0$$

which is equivalent to the stated result.

4. • If $H = 1$:

This is proven directly using (2.2.1), and by computing :

$$\text{Cov}(X_s, X_t) = \mathbb{E}[X_sX_t] - \mathbb{E}[X_s]\mathbb{E}[X_t]$$

using the H -self-similarity again.

• If $0 < H < 1$:

Let $s, t \in \mathbb{R}$. In this case, we just proved that $\mathbb{E}[X_s] = \mathbb{E}[X_t] = 0$.

Thus :

$$\begin{aligned} \text{Cov}(X_s, X_t) &= \mathbb{E}[X_sX_t] - \underbrace{\mathbb{E}[X_s]\mathbb{E}[X_t]}_{=0} \\ &= \mathbb{E}[X_sX_t] \\ &= \frac{1}{2} (\mathbb{E}[X_s^2] + \mathbb{E}[X_t^2] - \mathbb{E}[(X_s - X_t)^2]) \end{aligned}$$

Now, since the process has stationary increments, and is H -self-similar :

$$\forall s, t \in \mathbb{R}, \quad (X_s - X_t)^2 \stackrel{(d)}{=} (X_{|s-t|} - X_0)^2 \stackrel{(d)}{=} X_{|s-t|}^2 \stackrel{(d)}{=} |s - t|^{2H} X_1^2$$

Thus, one has :

$$\begin{aligned} \text{Cov}(X_s, X_t) &= \frac{1}{2} (\mathbb{E}[(s^H X_1)^2] + \mathbb{E}[(t^H X_1)^2] - \mathbb{E}[|s - t|^{2H} X_1^2]) \\ &= \frac{\mathbb{E}[X_1^2]}{2} (|s|^{2H} + |t|^{2H} - |s - t|^{2H}) \end{aligned}$$

which ends the proof. □

2.3 Fractional Brownian Motion

In this section, we finally arrive at the definition of the object and discuss its statistical properties, which are influenced by the parameter H .

Definition 2.7 (Fractional Brownian Motion).

The *fractional Brownian motion* (fBm) with *Hurst exponent* $H \in (0, 1)$ is the unique centered Gaussian stochastic process, denoted $(B_t^H)_{t \in \mathbb{R}}$, with variance $\text{Var}(B_1^H) = \sigma_H^2 > 0$, that is H -self-similar, has stationary increments, and possesses almost surely continuous sample paths. When $\sigma_H^2 = 1$, the process is called a *standard fractional Brownian motion*. Moreover, if $H = 1/2$, the process reduces to a *Brownian motion*, which is called a *standard Brownian motion* (or *Wiener process*) when $\sigma_H^2 = 1$.

This process is indeed well defined, in part due to theorem 2.6, which shows that it is uniquely determined for a given $\sigma_H^2 > 0$, since it fixes the covariance through $\text{Var}(B_1^H) = \mathbb{E}[(B_1^H)^2]$. However, it amounts to showing that the process is Gaussian, which we do not address here for the sake of conciseness, as it would require some additional material.

Special case when $H = 1/2$ (Brownian motion) :

One important property to notice about fractional Brownian motion is that its increments $(B_s^H - B_t^H)_{s>t}$ are independent when $H = 1/2$, since the process is gaussian with $\text{Cov}(B_s^{1/2}, B_t^{1/2}) = 0$ (by theorem 2.6). Therefore, it precisely matches the definition of what is called a *Lévy stochastic process*. However, there is no reason for this property to hold when $H \neq 1/2$, and indeed it does not.

Non-stationarity of fBm :

Indeed, given that :

$$\forall H \in (0, 1), \forall t \in \mathbb{R}, \quad \text{Var}(B_t^H) = \text{Cov}(B_t^H, B_t^H) = \sigma_H^2 |t|^{2H}$$

one observes that the distribution of B_t^H changes over time (even for Brownian motion). Hence, fractional Brownian motion is a non-stationary process for any Hurst exponent $H \in (0, 1)$.

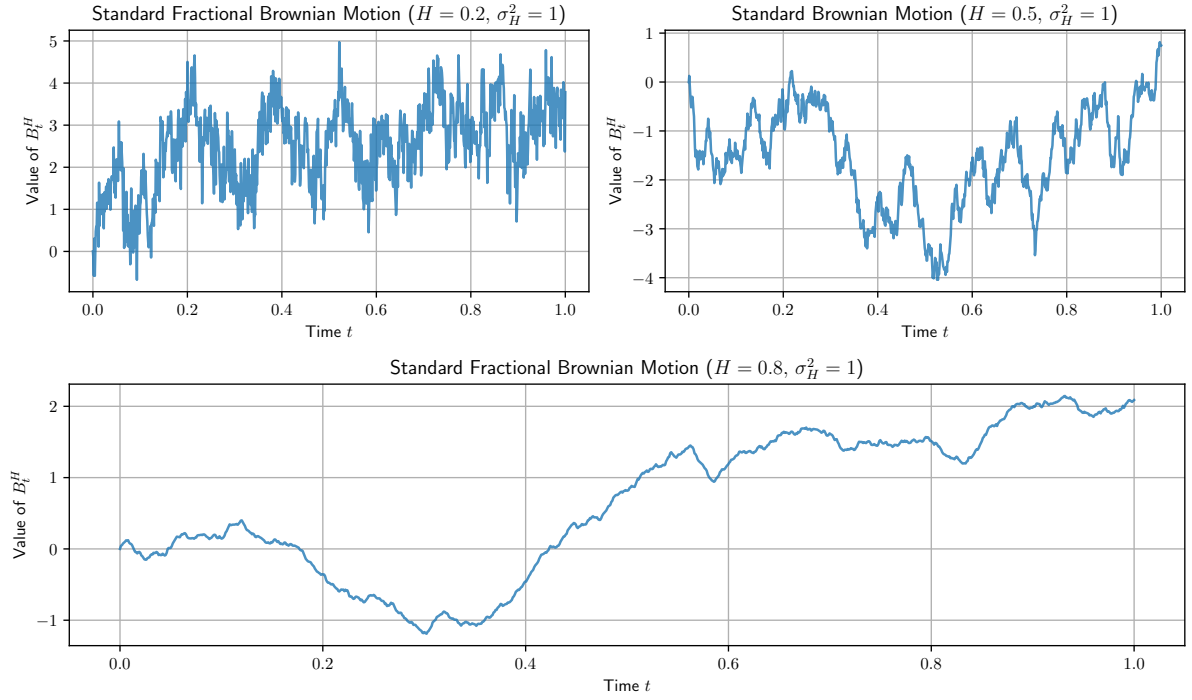


Figure 3: **Realization of three standard fBms** with Hurst exponents $H = 0.2$, $H = 0.5$, and $H = 0.8$.

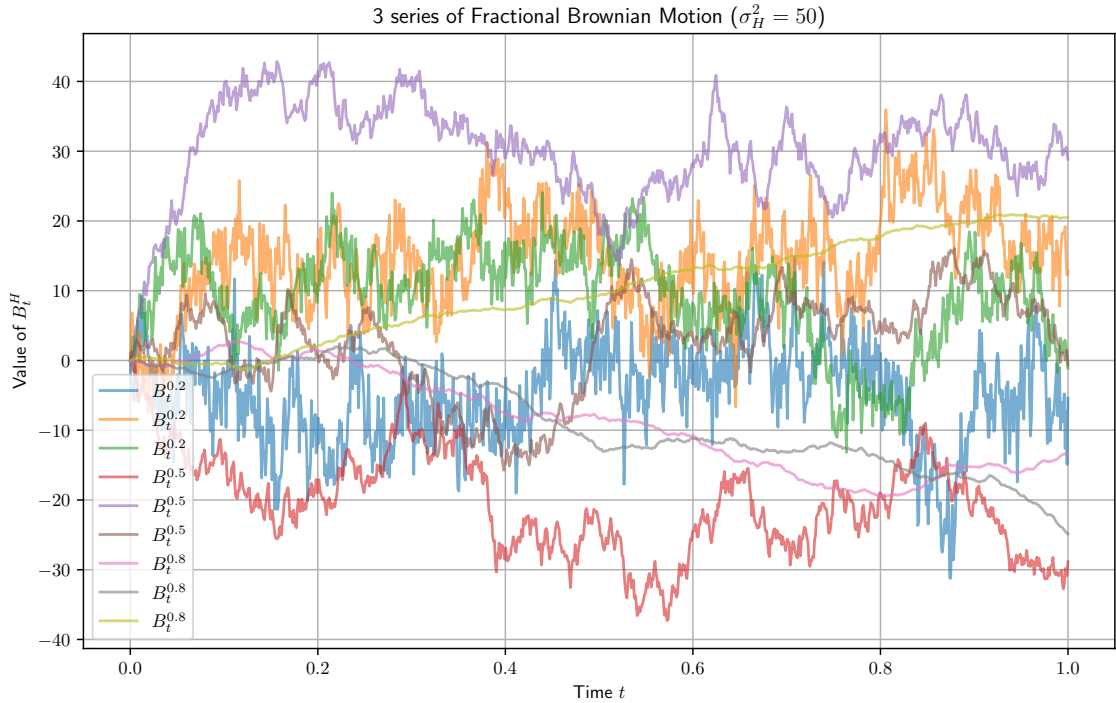


Figure 4: **Independent realizations of three fBms** with variance $\sigma_H^2 = 50$, for each Hurst exponent $H = 0.2$, $H = 0.5$, and $H = 0.8$.

2.4 The spectrum of a stochastic process

2.4.1 Spectrum of a stationary stochastic process

In signal processing and physics, the frequency content of a signal is often more informative than its time-domain representation. For deterministic signals, the Fourier transform provides direct access to this information. However, for stochastic processes, a single deterministic realization is not sufficient to characterize the process. Instead, one must rely on statistical properties and average behaviors, which motivates the introduction of a spectral notion describing how variance or power is distributed across frequencies.

Directly applying the Fourier transform to a process $(X_t)_{t \in \mathbb{R}}$ generally fails as :

- Sample paths may not belong to $L^2(\mathbb{R})$.
- The resulting transform would be random, whereas we seek a deterministic quantity characterizing the stochastic process itself.

To define the spectrum of a stochastic process, we therefore need a more regular object than the Fourier transform of a single realization. One common approach consists in truncating the signal over a finite interval $[-T, T]$, computing the Fourier transform of this truncated signal, and then considering its average energy, defined as the squared magnitude of the transform divided by $2T$. To remove the dependence on the truncation interval, we take the limit of this quantity as $T \rightarrow +\infty$. This limit, when it exists, defines the power spectral density (PSD) which we define now.

Definition 2.8 (Power Spectral Density (PSD)).

If we call $\tilde{S}$ the power spectral density (PSD), then :

$$\forall \omega \in \mathbb{R}, \quad \tilde{S}(\omega) := \lim_{T \rightarrow +\infty} \left(\frac{|\mathcal{F}[X\mathbf{1}_{[-T, T]}](\omega)|^2}{2T} \right) := \lim_{T \rightarrow +\infty} \frac{1}{2T} \left| \int_{-T}^T X_t e^{-2i\pi\omega t} dt \right|^2$$

Nevertheless, this construction still yields a random quantity, since it is defined from the stochastic process itself and therefore depends on the underlying randomness. However, our objective is to obtain a unique, deterministic object that characterizes the stochastic process as a whole, rather than a random outcome. This can be achieved by taking the expectation of the previously defined quantity, which captures the average behavior of the process.

Definition 2.9 (Power Spectrum Density of a stochastic process).

The power spectral density of a stationary stochastic process $(X_t)_{t \in \mathbb{R}}$ is defined by :

$$\forall \omega \in \mathbb{R}, \quad S_{XX}(\omega) := \lim_{T \rightarrow +\infty} \mathbb{E} \left[\frac{|\mathcal{F}[X\mathbf{1}_{[-T, T]}](\omega)|^2}{2T} \right] := \lim_{T \rightarrow +\infty} \frac{1}{2T} \mathbb{E} \left[\left| \int_{-T}^T X_t e^{-2i\pi\omega t} dt \right|^2 \right]$$

We have defined the spectrum of a stationary stochastic process through its power spectral density (PSD). Within this framework, there exists a fundamental relationship between the PSD and a quantity known as the autocorrelation function.

Definition 2.10 (Autocovariance).

For a stochastic process $(X_t)_{t \in \mathbb{R}}$ with finite second order moments, its autocovariance $K_{XX}(t_1, t_2)$ at time $t_1, t_2 \in \mathbb{R}$ is defined as :

$$K_{XX}(t_1, t_2) := \mathbb{E} \left[(X_{t_1} - \mathbb{E}[X_{t_1}]) (X_{t_2} - \mathbb{E}[X_{t_2}]) \right]$$

Definition 2.11 (Autocorrelation).

The autocorrelation function of a stochastic process $(X_t)_{t \in \mathbb{R}}$ at time $t_1, t_2 \in \mathbb{R}$, denoted R_{XX} , is defined by :

$$R_{XX}(t_1, t_2) := \mathbb{E}[X_{t_1} X_{t_2}]$$

Remark 1. The autocorrelation and autocovariance functions are related by :

$$\forall t_1, t_2 \in \mathbb{R}, \quad K_{XX}(t_1, t_2) = R_{XX}(t_1, t_2) - \mathbb{E}[X_{t_1}] \mathbb{E}[X_{t_2}]$$

2. In the stationary case, these definitions simplify to :

$$\forall \tau \in \mathbb{R}, \quad K_{XX}(\tau) := \mathbb{E}[X_t X_{t+\tau}] - \mathbb{E}^2[X_t] \quad \text{and} \quad R_{XX}(\tau) := \mathbb{E}[X_t X_{t+\tau}]$$

defined independently of the chosen time $t \in \mathbb{R}$.

Example Consider the following stationary stochastic process :

$$\forall t \in \mathbb{R}, \quad X_t \sim \mathcal{N}(0, 1),$$

where the random variables $\{X_t\}_{t \in \mathbb{R}}$ are independent and identically distributed.

The autocorrelation function is then given by :

$$\forall \tau \in \mathbb{R}, \quad R_{XX}(\tau) = \begin{cases} \mathbb{E}[X_t X_{t+\tau}] = \mathbb{E}[X_t] \mathbb{E}[X_{t+\tau}] = 0 & \text{if } \tau \neq 0, \\ \mathbb{E}[X_t^2] = 1 & \text{if } \tau = 0 \end{cases}$$

whatever $t \in \mathbb{R}$.

For a fractional Brownian motion, the autocorrelation function is illustrated in Figure 5.

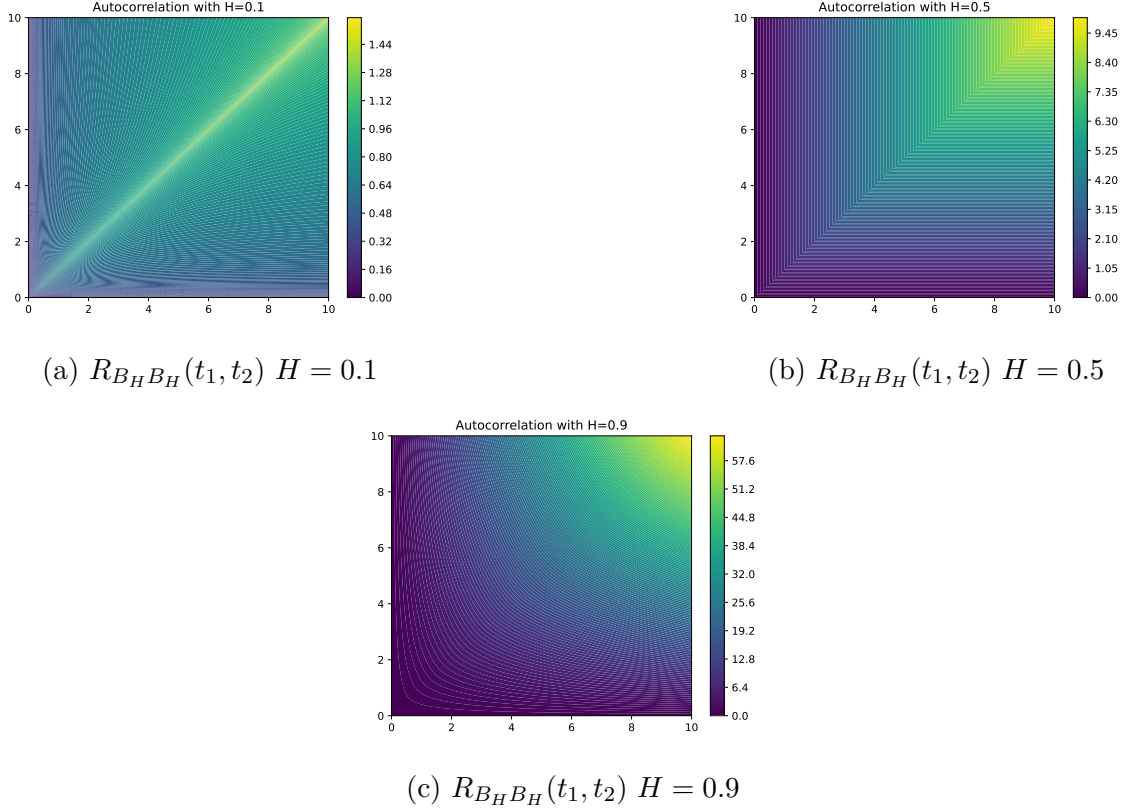


Figure 5: Autocorrelation of FBM processes

Here, the autocorrelation function illustrates the dependence of the signal with a time-shifted version of itself, which makes it possible to extract recurring "patterns" within the signal. In the first Gaussian example, since all random variables are independent, the correlation is zero for all nonzero time lags and reaches its maximum at zero lag, where the signal is perfectly correlated with itself.

Theorem 2.12 (Wiener–Khinchin).

If a stochastic process $(X_t)_{t \in \mathbb{R}}$ is stationary and of order 2 (i.e. admits second order moments), then :

$$\forall \omega \in \mathbb{R}, \quad S_{XX}(\omega) = \mathcal{F}[R_{XX}](\omega) := \int_{\mathbb{R}} \mathbb{E}[X_t X_{t+\tau}] e^{-2i\pi\omega\tau} d\tau$$

whatever $t \in \mathbb{R}$.

2.4.2 Spectrum of non-stationary processes

The spectrum of non-stationary processes is more difficult to define. Indeed, spectral analysis is fundamentally rooted in the assumption of stationarity. In the non-stationary setting, the spectrum must therefore depend on both time and frequency. Unfortunately, it is not possible to preserve all the properties of the stationary case when extending spectral notions to non-stationary processes. As a result, one must select which properties

to retain when constructing a spectral representation, implying that such a definition is not unique. Our objective is thus to propose a "reasonable" definition of the spectrum for "reasonable" non-stationary processes.

To this end, we introduce a class of non-stationary processes, called "harmonizable processes", for which a spectral representation can be defined.

Definition 2.13 (Harmonizable process).

A stochastic process $(X_t)_{t \in \mathbb{R}}$ is said to be *harmonizable* if there exists $g \in L^2(\mathbb{R}^2)$ such that :

$$\forall \omega_1, \omega_2 \in \mathbb{R}, \quad R_{XX}(t_1, t_2) = \int_{\mathbb{R}^2} g(\omega_1, \omega_2) e^{2i\pi(t_1\omega_1 - t_2\omega_2)} d\omega_1 d\omega_2$$

From this definition, one may attempt to construct a spectral representation satisfying a number of desirable properties, as discussed by *Loynes* in [10]. Unfortunately, in the non-stationary setting, some of these properties are mutually exclusive, which forces us to make specific choices in order to obtain a meaningful spectral representation.

1. We seek a linear relationship between the autocorrelation function and the spectrum. In the stationary case, this relationship is provided by the Wiener–Khinchin theorem (Theorem 2.12). In the non-stationary framework, this motivates spectral representations of the form proposed by *P.Flandrin* [8] :

$$S_{XX}(t, \omega) = \int_{\mathbb{R}^2} D(t, \omega, t_1, t_2) R_{XX}(t_1, t_2) dt_1 dt_2$$

2. We further require the spectrum to be invariant under time translations. Under this constraint, the previous expression can be reduced to the following form, as shown in the works of Flandrin and earlier contributions by *B.Escudié* and *J.Gréa* [8, 9, 11] :

$$S_{XX}(\omega, f) = \int_{\mathbb{R}^2} F(u - t, \tau) R_{XX}\left(u + \frac{\tau}{2}, u - \frac{\tau}{2}\right) e^{-2i\pi\omega\tau} du d\tau$$

3. We assume that the spectrum is real-valued and that the following diagram is commutative:

$$\begin{array}{ccc} R_{XX}(t_1, t_2) & \begin{array}{c} \xrightarrow{\mathcal{F}} \\ \xleftarrow{\mathcal{F}^{-1}} \end{array} & \mathcal{F}(R_{XX})(\omega_1, \omega_2) \\ & \searrow \quad \swarrow & \\ & S_{XX}(t, \omega) & \end{array}$$

Under these assumptions, the spectral representation becomes unique. The resulting time–frequency spectrum, leads to the *Wigner–Ville spectrum*, known from [8, 9, 11], which we now define.

Definition 2.14 (Wigner–Ville spectral density).

The *Wiener–Ville spectrum* of a stochastic process $(X_t)_{t \in \mathbb{R}}$, denoted S_{XX} is defined by :

$$\forall (t, \omega) \in \mathbb{R}^2, \quad S_{XX}(t, \omega) := \int_{\mathbb{R}} R_{XX} \left(t + \frac{\tau}{2}, t - \frac{\tau}{2} \right) e^{-2i\pi\omega\tau} d\tau$$

Remark The Wiener–Ville spectral density (WVSD) is the unique linear transformation of the autocorrelation function that is invariant under time translations and yields real-valued results, while extending the classical power spectral density (PSD) [8, 9]. This representation preserves the interpretation of the spectrum as a ”frequency density”, at the cost of allowing negative values. In the stationary case, this definition reduces to the standard PSD.

It is possible to remove the explicit time dependence by considering the averaged Wiener–Ville spectrum (AWVSD).

Definition 2.15 (Averaged Wiener–Ville spectral density (AWVSD)).

The *averaged Wiener–Ville spectral density* of a stochastic process $(X_t)_{t \in \mathbb{R}}$ is defined as :

$$\forall \omega \in \mathbb{R}, \quad \mathcal{S}(\omega) := \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T S_{XX}(t, \omega) dt$$

In the special case of an FBM, we obtain:

Theorem 2.16 (WVSD of fractional Brownian motion).

The *Wigner–Ville spectral density* of a given fBm $(B_t^H)_{t \in \mathbb{R}}$ with Hurst exponent H reads as :

$$\forall (t, \omega) \in \mathbb{R}^2, \quad S_{B_H B_H}(t, \omega) := \frac{1 - 2^{1-2H} \cos(4\pi\omega t)}{|2\pi\omega|^{2H+1}}$$

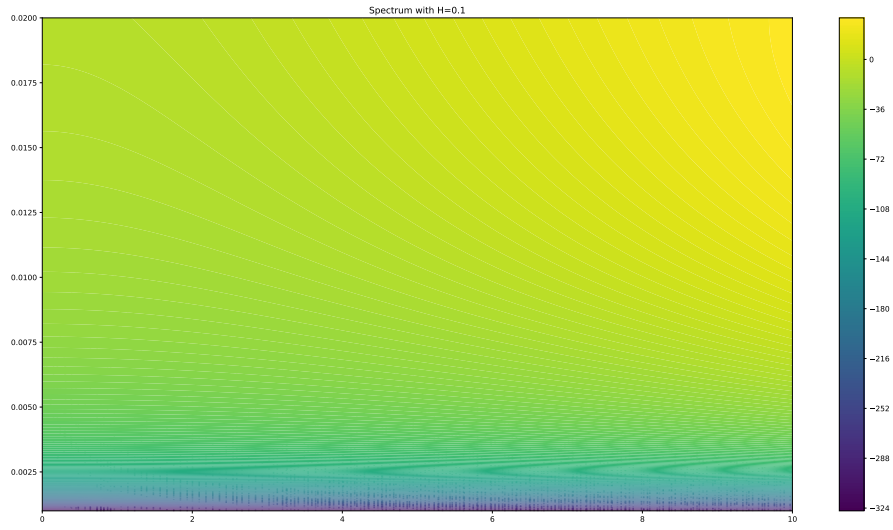


Figure 6: $S_{B_H B_H}(t, f)$ $H = 0.1$

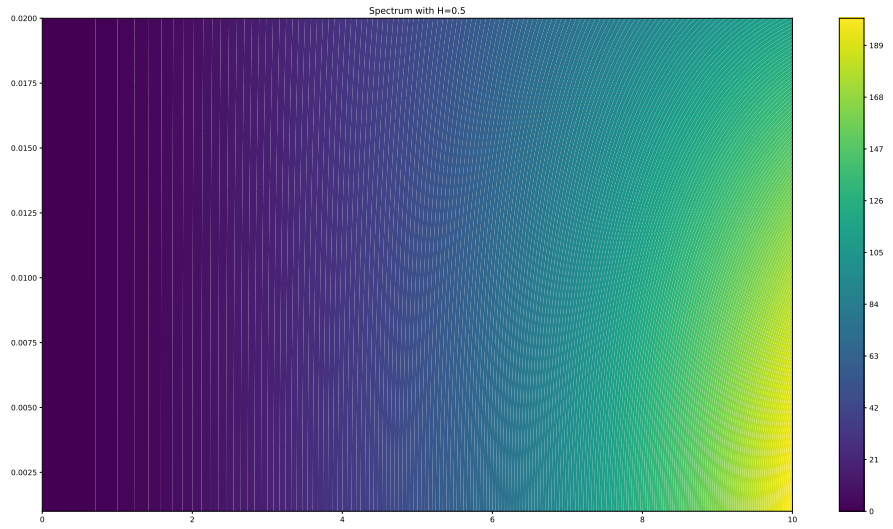


Figure 7: $S_{B_H B_H}(t, f)$ $H = 0.5$

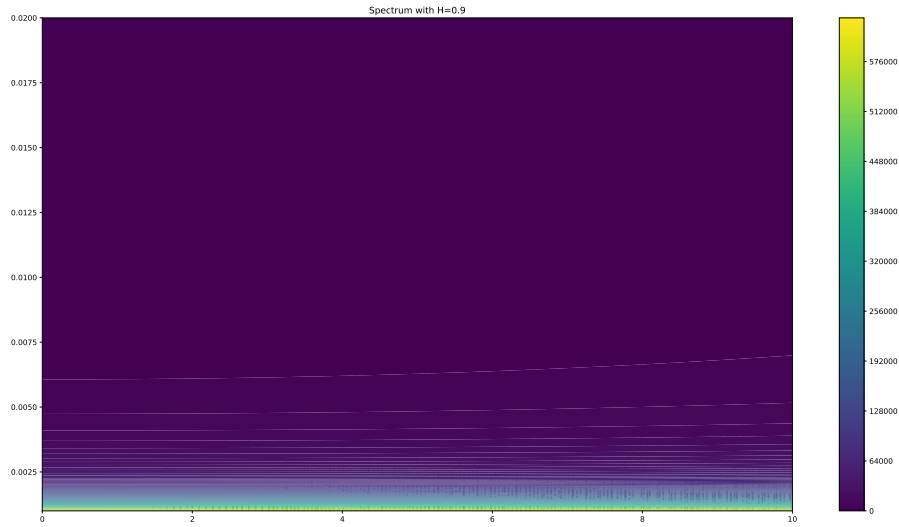


Figure 8: $S_{B_H B_H}(t, f)$ $H = 0.9$

Some properties of the FBM can be retrieved from the spectrum:

Theorem 2.17 (Self-Similarity of fractional Brownian motion in WVSD).

Let $(B_t^H)_{t \in \mathbb{R}}$ be a fBm with Hurst exponent H , and $\varepsilon > 0$. Then, denoting $\mathcal{E}_t := B_{\varepsilon t}^H$, one has that :

$$S_{\mathcal{E}\mathcal{E}}(t, \omega) = \frac{S_{B^H B^H}(\varepsilon t, \omega/\varepsilon)}{\varepsilon} = S_{(\varepsilon^H B^H)(\varepsilon^H B^H)}(t, \omega)$$

Theorem 2.18 (AWVSD of fractional Brownian motion).

The averaged Wigner–Ville spectral density of a fBm $(B_t^H)_{t \in \mathbb{R}}$ with Hurst exponent H and variance σ_H^2 is given by :

$$\forall \omega \in \mathbb{R}, \quad \mathcal{S}_{B_H}(\omega) = \frac{\sigma_H^2}{|2\pi\omega|^{2H+1}}$$

As an immediate consequence, fractional Brownian motion, regardless of the value of the Hurst parameter, is classified as a stochastic process exhibiting a $1/f$ spectrum. In what follows, and in particular in Section 4.2, this spectral form will be used throughout, in accordance with the approach adopted in the reference study by *P.Flandrin* [1].

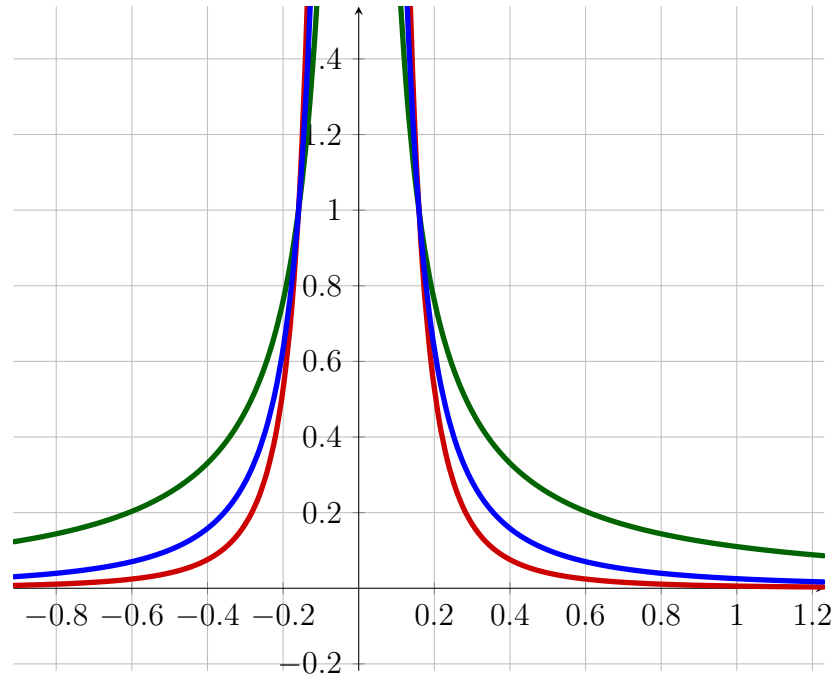


Figure 9: Averaged Wigner–Ville spectral density $\mathcal{S}$ of a fBm

- (a) Green : $H = 0.1$
- (b) Blue : $H = 0.5$
- (c) Red : $H = 0.9$

If you are intrested in the proof of the theorem above you can look at: [[9](#), [8](#), [7](#), [10](#), [11](#)]

3 A Wavelet Analysis of Fractional Brownian Motions

As we have just seen, the two main intrinsic properties that characterize fractional Brownian motion are its self-similarity and non-stationarity. As discussed in section 2.1, self-similarity captures the idea that the process exhibits structural patterns across all scales. On the other hand, non-stationarity reflects the fact that the statistical properties of the process evolve over time. This naturally highlights a duality between scale and time information. Consequently, in order to simultaneously capture both scale-dependent and temporal features, wavelets appear to be a particularly well-suited tool, as they are inherently designed to analyze signals through a trade-off between time and scale localization.

3.1 Statistical properties of fBm wavelet coefficients

To carry out the analysis, we focus on applying a discrete wavelet transform (DWT) based on an orthonormal basis :

$$\{\psi_{j,n} := 2^{-j/2} \psi(2^{-j}t - n) \mid (j, n) \in \mathbb{Z}^2\}$$

constructed from a given mother wavelet ψ . We further denote by $J \in \mathbb{Z}$ the index corresponding to the frequency-scale resolution 2^J , which defines the separation between low and high dyadic scales (i.e., between low and high frequencies). Recall that the low-frequency content (i.e., $\{2^j < 2^J \mid j \in \mathbb{Z}\}$) is encoded by a low-pass filter, namely the scaling function ϕ . Since the sample paths are almost surely continuous, the discrete wavelet transform of a fractional Brownian motion can be computed using this dictionary.

Thus, by applying the DWT on a fBm, we end up having :

- Approximation coefficients :

$$a_j[n] := \int_{\mathbb{R}} B_t^H \phi_{j,n}(t) dt = 2^{-j/2} \int_{\mathbb{R}} B_t^H \phi(2^{-j}t - n) dt \quad j, n \in \mathbb{Z}$$

- Detail coefficients :

$$d_j[n] := \int_{\mathbb{R}} B_t^H \psi_{j,n}(t) dt = 2^{-j/2} \int_{\mathbb{R}} B_t^H \psi(2^{-j}t - n) dt \quad j, n \in \mathbb{Z}$$

As a result, and unsurprisingly, the wavelet coefficients of a fBm are random variables at each scale 2^j and time index n . In fact, both the scale-indexed sequences $(a_j[n])_{j \in \mathbb{Z}}$ and $(d_j[n])_{j \in \mathbb{Z}}$, as well as the time-indexed sequences $(a_j[n])_{n \in \mathbb{Z}}$ and $(d_j[n])_{n \in \mathbb{Z}}$, can be viewed

as discrete-time stochastic processes, which decompose fBm almost surely as follows :

$$\forall t \in \mathbb{R}, \quad B_t^H = 2^{-J/2} \sum_{n=-\infty}^{+\infty} a_J[n] \phi(2^{-j}t - n) + \sum_{j=-\infty}^J 2^{-j} \sum_{n=-\infty}^{+\infty} d_j[n] \psi(2^{-j}t - n) \quad \text{a.s.}$$

for a given scale resolution 2^J .

This observation motivates the study of their statistical properties in order to establish a link with the underlying stochastic process $(B_t^H)_{t \in \mathbb{R}}$ itself.

Theorem 3.1 (Correlation between detail fBm wavelet coefficients).

Let $(B_t^H)_{t \in \mathbb{R}}$ be a fBm with Hurst exponent H . Then, for any $(j, n), (k, m) \in \mathbb{Z}^2$, the correlation between its detail wavelet coefficients can be expressed in the following way :

$$\mathbb{E}[d_j[n]d_k[m]] = \frac{\sigma_H^2}{2} \left(- \int_{\mathbb{R}} A_\psi(2^{j-k}, \tau - (2^{j-k}n - m)) |\tau|^{2H} d\tau \right) (2^k)^{2H+1}$$

$$\text{where } A_\psi : (x, y) \in \mathbb{R}^2 \mapsto \sqrt{x} \int_{\mathbb{R}} \psi(t) \psi(xt - y) dt \in \mathbb{R}$$

This further implies that, for a given $j \in \mathbb{Z}$, the sequence $(d_j[n])_{n \in \mathbb{Z}}$ is stationary in the sense that the correlation depends only on the time shift $n - m$. Similarly, for a given $n \in \mathbb{Z}$, the sequence $(d_j[n])_{j \in \mathbb{Z}}$ is stationary in the sense that the correlation between $d_j[n]$ and $d_k[2^{j-k}n]$ depends only on the scale shift $j - k$.

Proof. The sketch is provided in the reference article [1]. □

Remark One recognizes A_ψ as the wavelet transform of the mother wavelet ψ . This mapping is also referred to as the *wide-band ambiguity function* and can essentially be interpreted as a time-scale correlation of the wavelet dictionary with itself.

What is striking here is that, although we have recovered the expected self-similarity and stationarity across scales from the fBm, the detail coefficients appear to be stationary at any given scale. This is very surprising, given that fBm itself is known to be non-stationary, as discussed in Section 2.3.

However, the reason for this is actually explained in the proof of theorem 3.1. Indeed, to establish the result, we used the fact that ψ is an admissible wavelet and therefore has zero mean, i.e., $\int_{\mathbb{R}} \psi = 0$. From this, by letting the resolution tend to infinity, one can conclude that the information responsible for the fBm's non-stationarity is concentrated very locally at the frequency $\omega = 0$, and is consequently not captured by the details.

In other words, the information responsible for the non-stationarity is expected to be encoded primarily within the approximation space, through the approximation coefficients, and at very low frequency scales. We now turn our attention to the statistical properties of these coefficients.

Theorem 3.2 (Variance of fBm wavelet coefficients).

Let $(B_t^H)_{t \in \mathbb{R}}$ be a fBm with Hurst exponent H . Then, for any $(j, n) \in \mathbb{Z}^2$, the fBm wavelet coefficients have following variance :

• **Approximation** :

$$\text{Var}(a_j[n]) = \frac{\sigma_H^2}{2} \left(- \int_{\mathbb{R}} (\gamma_\phi(\tau) - 2\phi(\tau - n)) |\tau|^{2H} d\tau \right) (2^j)^{2H+1} \quad (3.1.1)$$

• **Details** :

$$\text{Var}(d_j[n]) = \frac{\sigma_H^2}{2} \underbrace{\left(- \int_{\mathbb{R}} \gamma_\psi(\tau) |\tau|^{2H} d\tau \right)}_{=: V_\psi(H)} (2^j)^{2H+1} \quad (3.1.2)$$

Proof. The sketch is again provided in the reference article [1]. $\square$

Indeed, one observes that the approximation coefficients, although they are also self-similar, are non-stationary, since their variance, given by (3.1.1), depends on the time index n .

3.2 Log-Scale Estimation of the Hurst Exponent

3.2.1 DWT-Based Estimation

One may have noticed that, for a fixed time index $n \in \mathbb{N}$, the variance of the detail coefficients as a function of the scale 2^j follows a dyadic power law, since $V_\psi(H)$ from (3.1.2) does not depend on the scale. Thus, one can construct an estimator $\widehat{H}$ of the Hurst exponent by taking the slope of (3.1.2) in dyadic log-log scale since :

$$\begin{aligned} \forall (j, n) \in \mathbb{Z}^2, \quad \log_2 \text{Var}(d_j[n]) &= (2H + 1)j + c, \quad c \in \mathbb{R} \\ \implies \widehat{H} &:= \frac{\text{slope}(\log_2 \text{Var}(d_j[n])) - 1}{2} \end{aligned}$$

As discussed previously, although fractional Brownian motion is not a stationary process, its wavelet detail coefficients are stationary over time. This property plays a crucial role here, as it allows us to compute an empirical estimate of the variance and, consequently, to estimate the Hurst parameter. Hence, for a fixed scale 2^j , we are able to compute the empirical variance for N samples :

$$\overline{d_j[\cdot]} := \frac{1}{N} \sum_{n=0}^{N-1} |d_j[n]|^2 \quad (\text{zero mean process})$$

As we can see from table 1 and figure 10, the estimation performs well overall. However, the estimator becomes biased for small values of H , occasionally yielding negative

Table 1: **Hurst estimation results**

H	0.1	0.2	0.3	0.4	0.6	0.7	0.8	0.9
$\hat{H}$	-0.05941	0.1545	0.2616	0.3688	0.6287	0.656	0.7215	0.9014
$ H - \hat{H} $	0.1594	0.04555	0.03843	0.03123	0.02874	0.04399	0.07851	0.001391

estimates as shown further in figure 11. This behavior will be explained later in the upcoming section.

3.2.2 CWT-Based Estimation

In order to further illustrate the importance of separating detail and approximation scales, we consider constructing the same estimator using a *Continuous Wavelet Transform* (CWT), defined by

$$W(a, b) := \frac{1}{\sqrt{a}} \int_{\mathbb{R}} B_t^H \psi\left(\frac{t-b}{a}\right) dt, \quad (a, b) \in \mathbb{R}_+ \times \mathbb{R},$$

and restricting ourselves to dyadic scales $a = 2^j$ for ease of comparison with the DWT-based approach.

As a result, figures 12 and 13 show that the estimator exhibits a significant bias, especially for extreme values of the Hurst parameter.

Why? In contrast to the DWT, which explicitly separates the signal into approximation and detail components, the CWT produces coefficients that mix contributions from different frequency bands at each scale. More precisely, for a given scale a , the CWT coefficients :

$$W(a, b) := \frac{1}{\sqrt{a}} \int_{\mathbb{R}} B_t^H \psi\left(\frac{t-b}{a}\right) dt, \quad (a, b) \in \mathbb{R}_+ \times \mathbb{R},$$

contain both high- and low-frequency contributions, and are therefore not stationary in b . Consequently, the empirical variance :

$$\overline{W(a, \cdot)} = \frac{1}{N} \sum_b |W(a, b)|^2$$

is affected by the non-stationarity carried by the low-frequency components. This effect is particularly pronounced for large Hurst parameters ($H \approx 1$), where the fBm trajectories are very smooth and the high-frequency details are almost negligible, leading to a strong underestimation of the variance at fine scales and, hence, poor estimation performance.

However, for smaller values of H ($H \lesssim 0.5$), the high-frequency contributions asso-

ciated with small scales a dominate the empirical variance. Since the CWT coefficients satisfy the asymptotic scaling law

$$\mathbb{E} [|W(a, b)|^2] \underset{a \rightarrow 0}{\sim} C_\psi a^{2H+1}, \quad C_\psi > 0$$

the bias in the empirical variance estimator is reduced, and the resulting Hurst estimates are closer to those obtained with the DWT. Nevertheless, for very small Hurst exponents ($H \approx 0$), the extreme irregularity of the fBm trajectories induces large, highly fluctuating coefficients at fine scales, which again leads to biased and unstable estimates.

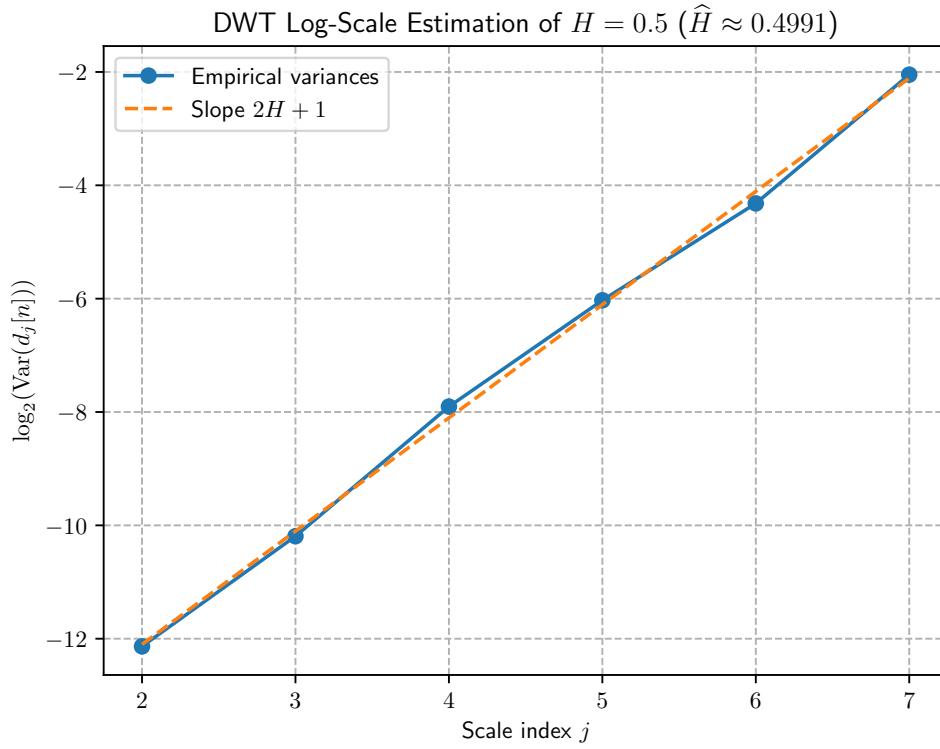


Figure 10: **DWT Log-Scale Estimation Result** on a standard fBm ($\sigma_H^2 = 1$).

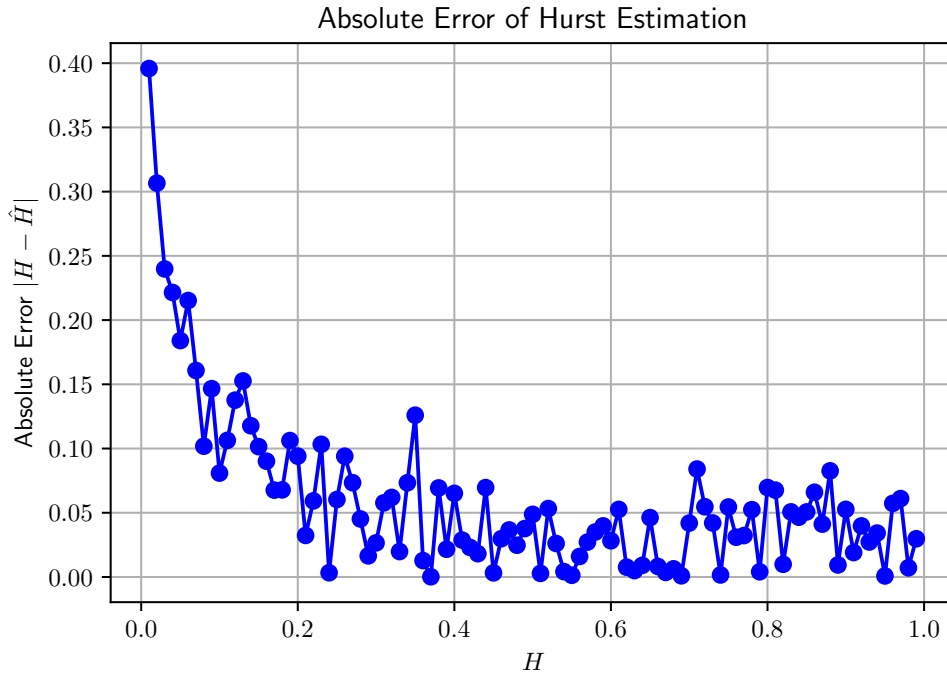


Figure 11: **Absolute error $|H - \hat{H}|$ for 100 uniformly distributed Hurst exponents**, each corresponding to a standard Brownian motion ($\sigma_H^2 = 1$).

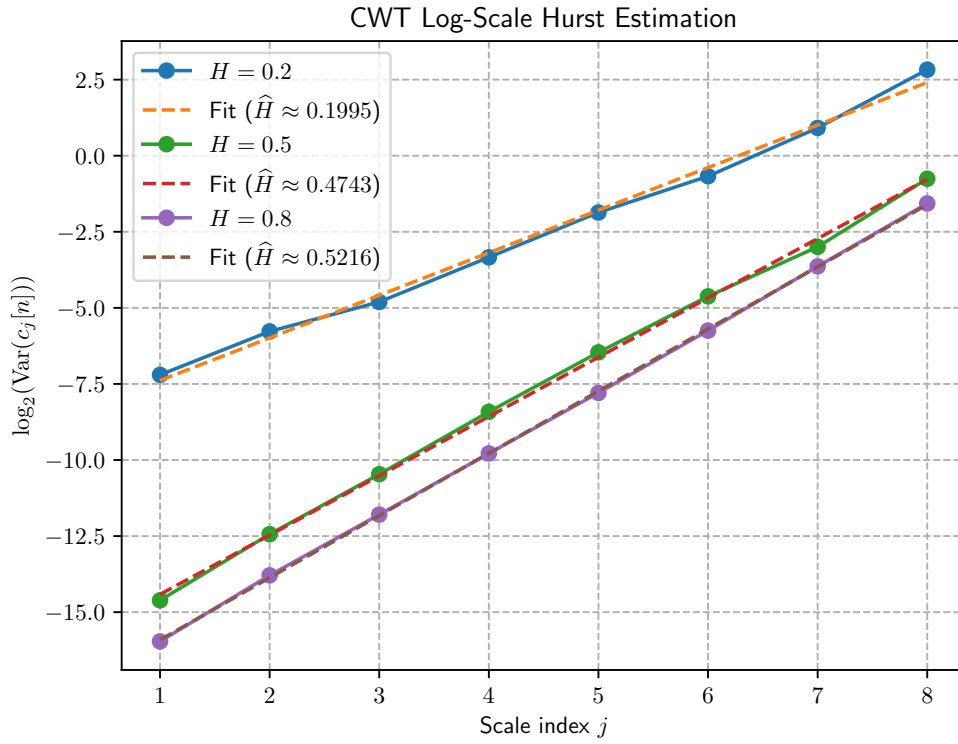


Figure 12: **CWT Log-Scale Estimation Result** on a standard fBm ($\sigma_H^2 = 1$) for $H = 0.2$, $H = 0.5$, and $H = 0.8$.

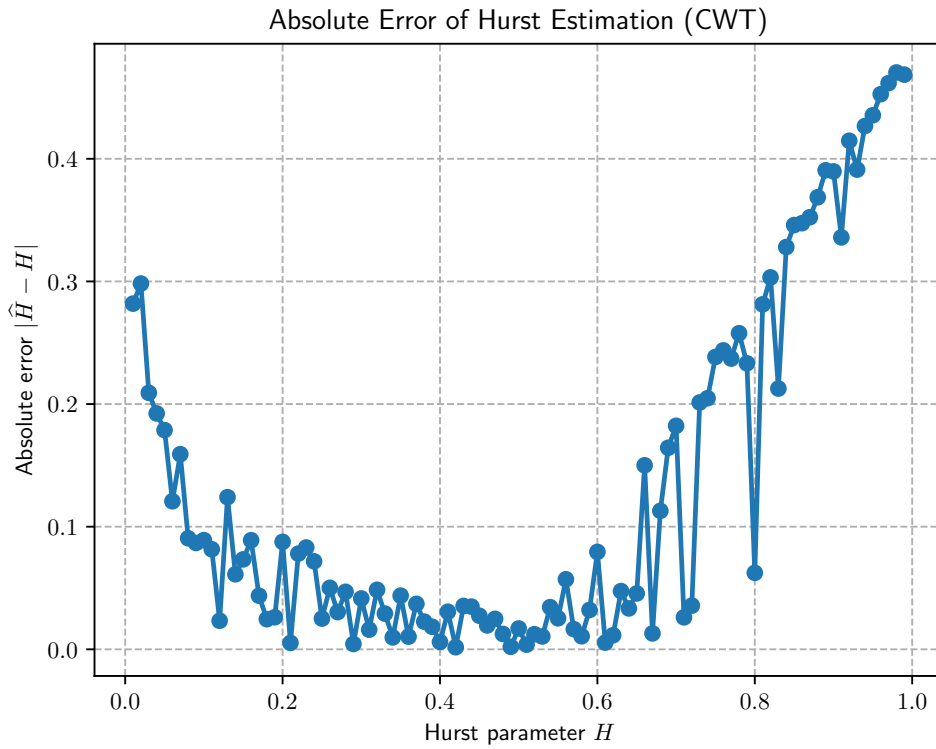


Figure 13: **Absolute error $|H - \hat{H}|$** using the CWT for 100 uniformly distributed Hurst parameters, each corresponding to a standard fBm ($\sigma_H^2 = 1$).

3.3 Asymptotical decay of correlation

Now that we have proved in section 3.1 that the wavelet coefficients of fBm are correlated both across scales and in time, we would like to further investigate the behaviour of these correlations, and how strong it can be. A natural question for instance is whether decorrelation can be achieved in some cases. In this section, we show that this depends strongly on the choice of the mother wavelet ψ , as well as, of course, on the Hurst exponent.

3.3.1 Case of the Haar Basis

As an illustration, we consider the example of the Haar basis, constructed from the mother wavelet :

$$\psi := \mathbf{1}_{[0, \frac{1}{2}]} - \mathbf{1}_{[\frac{1}{2}, 1]},$$

for which the correlation of fBm wavelet coefficients can be computed in closed form. This allows us to show that this basis exhibits very small correlation values, and even perfect decorrelation in certain settings.

Theorem 3.3 (Haar fBm wavelet correlation).

Let $(d_j[n])_{j,n \in \mathbb{Z}}$ be the Haar wavelet coefficients of a fBm with Hurst exponent H .

Then, for a given scale index $j \in \mathbb{Z}$, the time correlation reads :

$$\forall n, m \in \mathbb{Z}, \quad \mathbb{E}[d_j[n]d_j[m]] = \frac{\sigma_H^2}{2} \xi_H[n-m](2^j)^{2H+1} \quad (3.3.1)$$

$$\text{with} \quad \xi_H[k] := \begin{cases} \frac{1}{(2H+1)(2H+2)} \left[(|k|-1)^{2H+2} - 4(|k|-\frac{1}{2})^{2H+2} \right. \\ \quad \left. + 6|k|^{2H+2} - 4(|k|+\frac{1}{2})^{2H+2} + (|k|+1)^{2H+2} \right], & k \neq 0 \\ \frac{1-2^{-2H}}{(H+1)(2H+1)}, & k = 0 \end{cases}$$

for any $k \in \mathbb{Z}$.

Furthermore, if $H = 1/2$ (Brownian motion), then, given a fixed time-scale index $(n, j) \in \mathbb{Z}^2$, the fBm wavelet coefficients happen to be perfectly uncorrelated :

$$\mathbb{E}[d_j[n]d_k[m]] = 0$$

only outside of the cone-shaped time-scale domain defined for indices $(m, k) \in \mathbb{Z}^2$ verifying

$$2^{j-k}n \leq m \leq 2^{j-k}(n+1) - 1 \quad (3.3.2)$$

Haar wavelet correlation $\rho_j(n, m) := \mathbb{E}[d_j[n]d_j[m]]$ (normalized)

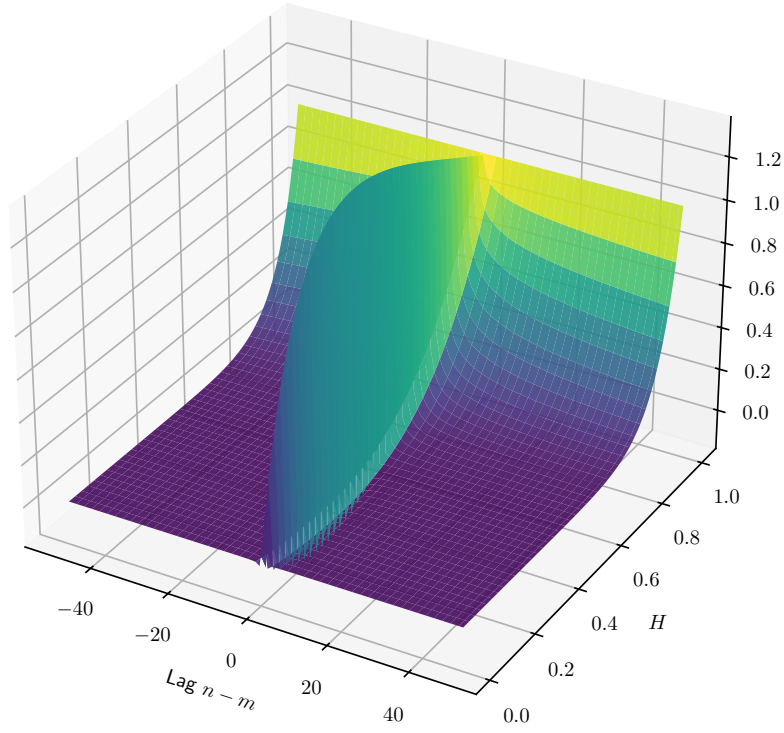


Figure 14: **Time Haar fBm wavelet correlation at scale $j = 0$** given by (3.3.1), depicting the asymptotic correlation decay.

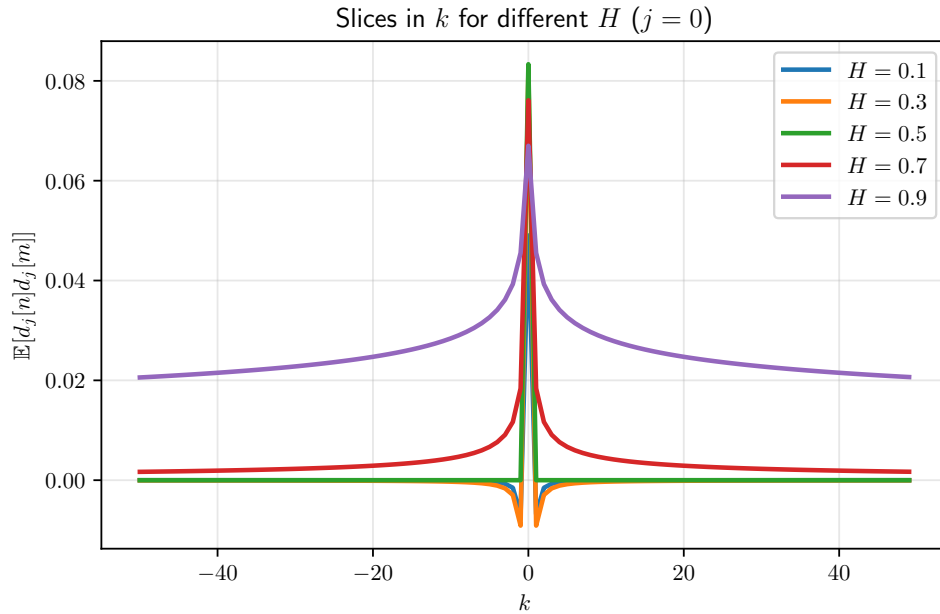


Figure 15: **Haar fBm wavelet correlation slices across lag k for multiple Hurst parameters**, showing how the correlation decays with lag k at scale $j = 0$.

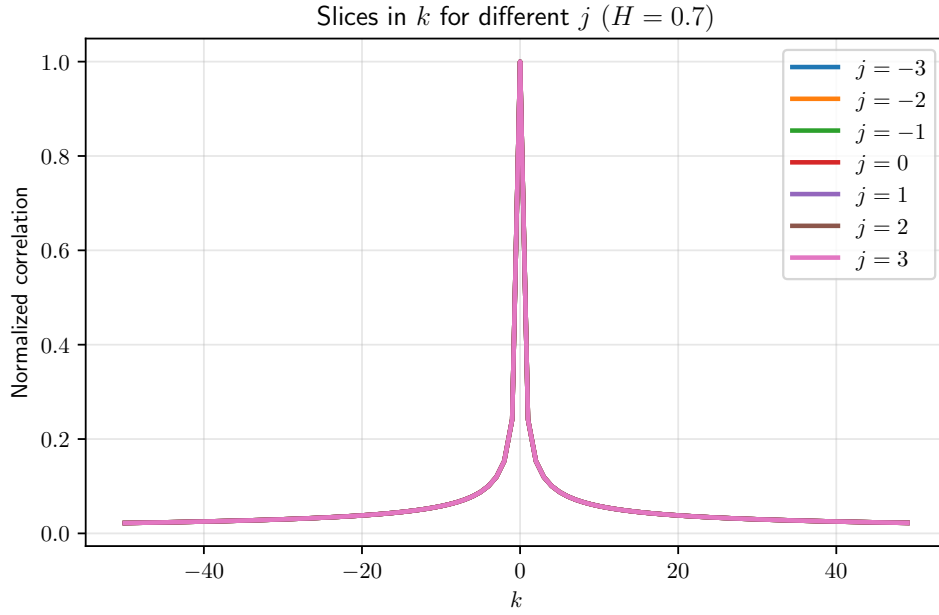


Figure 16: **Normalized Haar fBm wavelet correlation slices across lag k for multiple scales**, at fixed Hurst parameter $H = 0.7$, illustrating how the correlation shape changes with scale j .

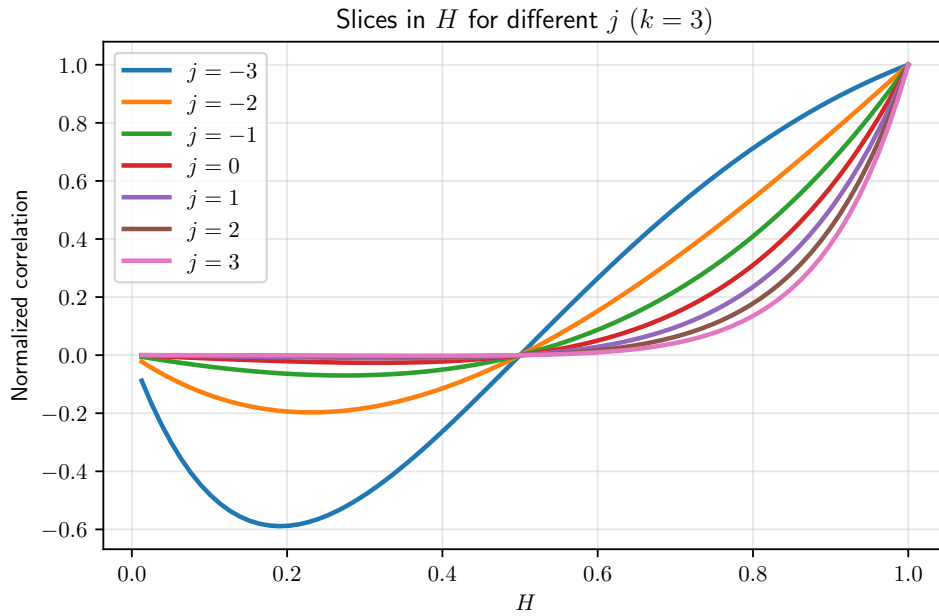


Figure 17: **Normalized Haar fBm wavelet correlation slices across Hurst parameter H for multiple scales**, at fixed lag $k = 3$, showing the effect of the Hurst parameter on correlation across scales.

From figures 14 and 15, one observes that the correlation converges to zero according to three distinct regimes :

- Fast decay : $0 < H < 1/2$,

- Perfect decorrelation : $H = 1/2$, achieved for every $m \neq n$, as the cone domain from (3.3.2) reduces to $\{(m, k) = (n, j)\}$ when $k = j$. More precisely, one can prove easily from (3.3.1) that :

$$\mathbb{E}[d_j[n]d_j[m]] = \delta_{n,m} \text{Var}(d_j[n]) = \delta_{n,m} \left(\frac{\sigma^2}{12} 2^{2j} \right)$$

- Long-range correlation (slow decay) : $1/2 < H < 1$. In fact, as mentioned in *P. Flandrin's* article [1], it is possible to show that the convergence to a near decorrelation behaves as :

$$\mathbb{E}[d_j[n]d_j[m]] \underset{|n-m| \rightarrow +\infty}{=} \mathcal{O}\left(|n-m|^{2(H-1)}\right) \quad (3.3.3)$$

Moreover, at a fixed Hurst exponent H , this decorrelation is completely unaffected and thus independent of the chosen scale, as can be seen more clearly in figure 16. However, as observed in figure 17, the scales do influence the initial correlation order, which tends to decrease very fast with the Hurst exponent as $2^j \geq 1$ becomes larger. On the other hand, very fine scales $2^j < 1$ give rise to larger initial orders (e.g., the blue curve corresponding to $j = -3$ in figure 17).

3.3.2 Vanishing moments

Having studied the asymptotic behaviour of the correlation with the Haar basis, we would now like to see whether it is possible to draw more general conclusions using a basis generated from a given mother wavelet ψ . Usually, the main property to consider is the number of vanishing moments the wavelet has, which we briefly recall here.

Definition 3.4 (Vanishing moments).

An admissible mother wavelet ψ is said to have R vanishing moments if its action against polynomials of degree $n \leq R - 1$ vanishes, i.e. :

$$\forall 0 \leq k \leq R - 1, \quad \int_{\mathbb{R}} t^k \psi(t) dt = 0$$

As we know from [4], wavelets with a high number of vanishing moments are well suited to capture local regularity information in signals, since they are able to cancel out the polynomial part of the Taylor expansion and thus emphasize the highest-order local variations of the signal. Furthermore, such wavelets tend to oscillate more as the number of vanishing moments increases, since they are required to be orthogonal to higher-order polynomials.

Now, the following proposition shows how these wavelets influence the decay of fBm

wavelet correlations, and in particular how the number of vanishing moments R of the chosen mother wavelet ψ impacts the rate at which correlations decrease over time.

Proposition 3.5 (Spectral form of fBm wavelet correlation).

Let $(d_j[n])_{j,n \in \mathbb{Z}}$ denote the wavelet coefficients of an fBm with Hurst exponent H , computed using a basis generated from a given mother wavelet ψ . Then, for a fixed scale index $j \in \mathbb{Z}$, the time correlation can be rewritten in a “spectral form”, in which the energy of the Fourier transform of ψ naturally appears:

$$\mathbb{E}[d_j[n] d_j[m]] = \frac{\sigma_H^2}{2} \left[\frac{\sin(\pi H) \Gamma(2H + 1)}{\pi} \left(\int_{\mathbb{R}} \frac{|\widehat{\psi}(\omega)|^2}{|\omega|^{2H+1}} e^{i\omega(n-m)} d\omega \right) \right] (2^j)^{2H+1}$$

Proof. □

From this point of view, it becomes clear that, in order for the correlation term to converge, the singular behaviour of $|\omega|^{-(2H+1)}$ at low frequencies must be compensated by the Fourier spectrum of the mother wavelet ψ . In other words, $\widehat{\psi}(\omega)$ has to vanish sufficiently fast as $\omega \rightarrow 0$. Assume that ψ has at most R vanishing moments.

Then one has :

$$\forall k \in \llbracket 0, R-1 \rrbracket, \quad \widehat{\psi}^{(k)}(0) = \int_{\mathbb{R}} t^k \psi(t) dt = 0$$

Since $\widehat{\psi}$ is $\mathcal{C}^\infty$, and in particular $\mathcal{C}^R$ in a neighbourhood V_0 of $\omega = 0$, the Taylor–Lagrange inequality applied at $\omega = 0$ yields the existence of a constant $C > 0$ such that :

$$\begin{aligned} \forall \omega \in V_0, \quad \left| \widehat{\psi}(\omega) \right| &= \left| \widehat{\psi}(\omega) - \sum_{k=0}^{R-1} \frac{\widehat{\psi}^{(k)}(0)}{k!} \omega^k \right| \leq C \frac{|\omega|^R}{R} \\ &\implies |\widehat{\psi}(\omega)| \underset{\omega \rightarrow 0}{=} \mathcal{O}(|\omega|^R) \end{aligned}$$

As a consequence,

$$\frac{|\widehat{\psi}(\omega)|^2}{|\omega|^{2H+1}} \underset{\omega \rightarrow 0}{=} \mathcal{O}(|\omega|^{2R-(2H+1)})$$

which is locally integrable around $\omega = 0$ as soon as :

$$2R - (2H + 1) \geq 0 \iff H \leq R - 1/2$$

which proves the following result.

Theorem 3.6 (Condition on vanishing moments for correlation decay).

Let $(d_j[n])_{j,n \in \mathbb{Z}}$ denote the wavelet coefficients of an fBm with Hurst exponent H , and let ψ be the associated mother wavelet having at most R vanishing moments. Then, correlation decay is ensured as soon as $H \leq R - 1/2$. Otherwise, if $H > R - 1/2$, one recovers a

long-range dependence (slow decay) with following rate :

$$\mathbb{E}[d_j[n] d_j[m]] \underset{|n-m| \rightarrow +\infty}{=} \mathcal{O}\left(|n-m|^{2(H-R)}\right).$$

One may notice that setting $R = 1$ recovers the decay rate (3.3.3) obtained with the Haar basis, since the Haar mother wavelet is only admissible and has a single vanishing moment. More generally, all admissible wavelets are thus shown to share the same asymptotic decay behaviour.

More importantly, as soon as mother wavelets ψ with at least two vanishing moments (i.e. $R \geq 2$) are considered, one obtains that :

$$H > R - \frac{1}{2} \implies H > 2 - \frac{1}{2} = \frac{3}{2} > 1,$$

which is impossible for a fractional Brownian motion. Hence, long-range correlations are ruled out as soon as wavelets with at least two vanishing moments are used.

Typical examples include *Daubechies* wavelets of order $N \geq 2$, *Symlets*, *Coiflets*, or more generally any compactly supported orthonormal wavelet with sufficiently many vanishing moments.

4 Synthesizing fBms through Wavelet Coefficients

Having studied the statistical properties of fractional Brownian motion (fBm) wavelet coefficients, we now turn to the question of synthesizing an fBm by generating wavelet coefficients with analogous statistical characteristics. This question is indeed quite natural, as Section 3.1 has shown how wavelet analysis provides deep insight into the self-similar structure of stochastic processes. It is therefore of interest to investigate whether one can construct similar families of processes by generating wavelet coefficients with appropriate scaling and statistical distribution.

Furthermore, as shown in Section 3.1, for a mother wavelet with a sufficient number of vanishing moments, the correlations between wavelet coefficients are negligible at all scales. This justifies the common approximation of treating the coefficients as independent Gaussian random variables with scale-dependent variance :

$$d_{j,k} \sim \mathcal{N}\left(0, \sigma_j^2\right), \quad \text{with} \quad \sigma_j^2 \propto 2^{-(2H+1)j}, \quad j = 1, \dots, J, \quad (4.0.1)$$

where $d_{j,n}$ denotes the wavelet coefficient at scale j and position n , and J is the maximum decomposition level. The coarsest approximation coefficient is typically set to zero (i.e., $c_A = 0$), in accordance with the property that $B_0^H = 0$ almost surely.

4.1 Inverse Discrete Wavelet Transform (IDWT)

A straightforward approach would be to reconstruct the trajectory using the IDWT applied to the generated coefficients $(d_j[n])_{n \in \mathbb{N}}$.

Using these independently generated coefficients, a full process can then be reconstructed via the inverse discrete wavelet transform (IDWT) :

$$\widetilde{B_{t_n}^H}^{(d)} := \text{IDWT}(c_A, d_J[n], d_{J-1}[n], \dots, d_1[n]) \quad (4.1.1)$$

where $(t_n)_n$ denotes the discrete dyadic time points.

As illustrated by the plots below, the simulated trajectories do not reproduce the characteristic structure of fractional Brownian motion (**at all**) and instead display noise-like behavior.

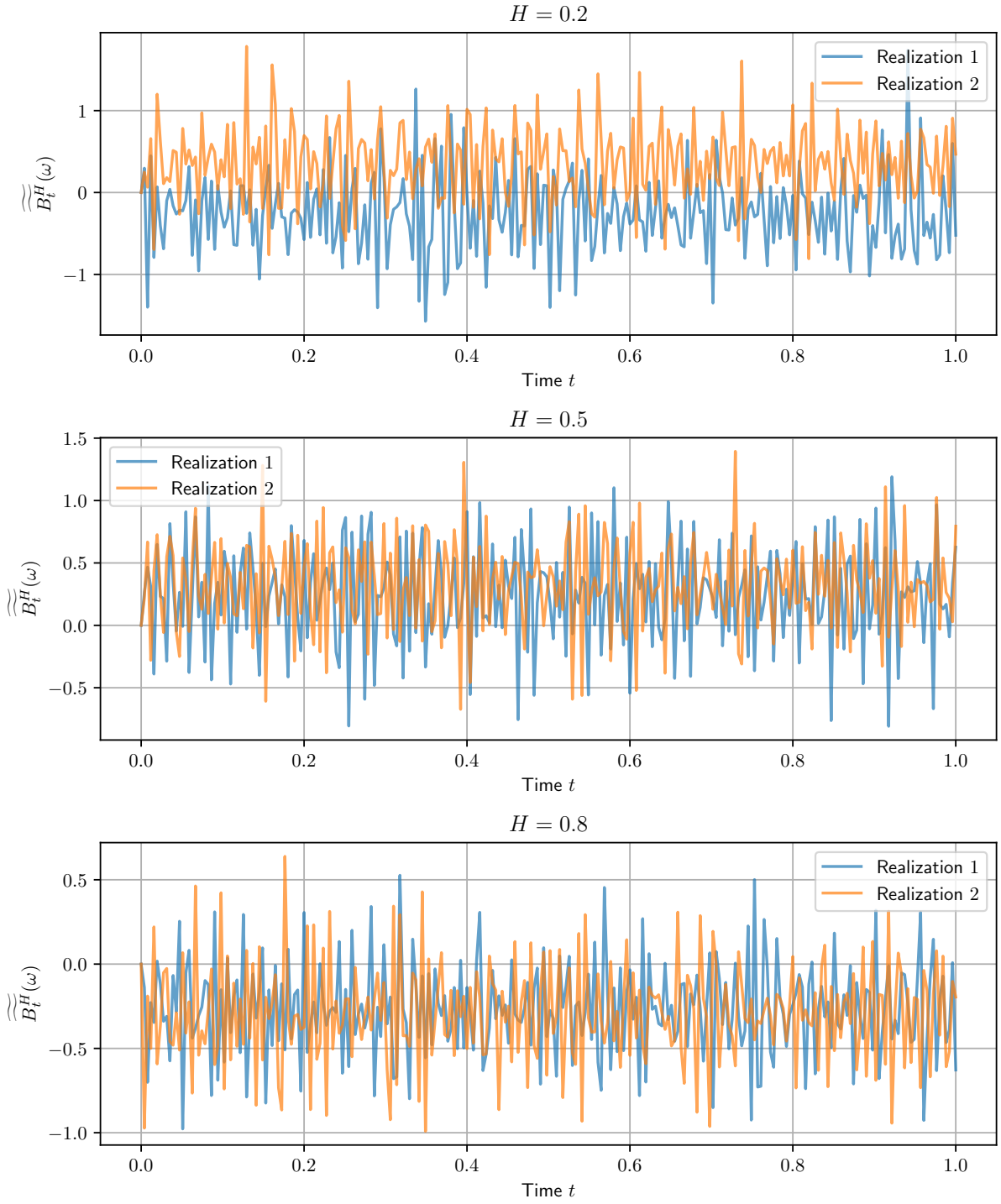


Figure 18: **DWT Log-Scale Estimation Result** on a standard fBm ($\sigma_H^2 = 1$).

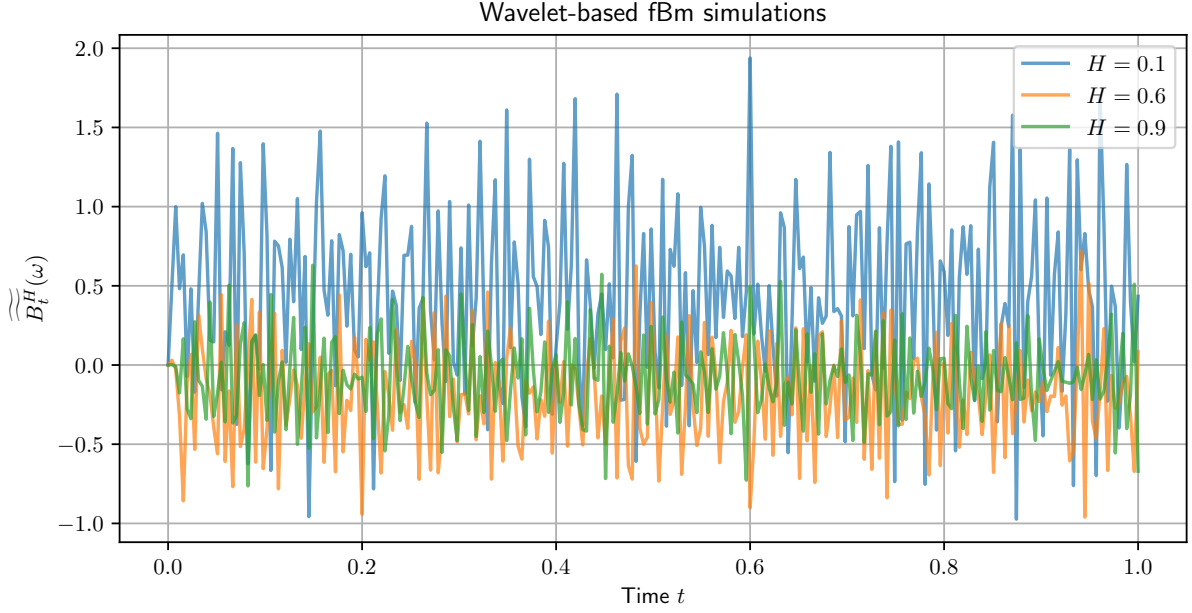


Figure 19: **DWT Log-Scale Estimation Result** on a standard fBm ($\sigma_H^2 = 1$).

4.2 Wornell's Orthonormal Synthesis

As discussed in Section 2.4.2, fractional Brownian motion with Hurst exponent H and variance σ_H^2 is partially characterized by a $1/f$ -type spectrum. More precisely, its averaged Wigner–Ville spectral density (AWVSD) $\mathcal{S}_{BH}$ reads :

$$\forall \omega \in \mathbb{R}, \quad \mathcal{S}_{BH}(\omega) = \frac{\sigma_H^2}{|\omega|^{2H+1}}$$

This motivates the reconstruction of processes from wavelet coefficients that would typically exhibit $1/f$ -type behavior as well, in order to obtain processes with characteristics similar to those of fractional Brownian motions.

In order to set a study framework for such non-stationary processes, *Gregory W. Wornell* proposed in his 1990 paper [6], an extension of the classical notion of spectrum. His approach, which relies on orthonormal wavelet expansions, aligns closely with the ideas we have been developing here. In fact, *G. Wornell* demonstrates that the uncorrelation assumption for fBm wavelet coefficients made in (4.0.1) is indeed justified, and shows that a family of $1/f$ -type processes can be generated. Consequently, his work leads to the following result.

Theorem 4.1 (G. Wornell, 1990).

Let $(d_j[n])_{j,n \in \mathbb{Z}}$ be the Gaussian random variables defined in (4.0.1), and let ψ be a mother wavelet associated to a orthonormal wavelet basis. Then, the process :

$$\widetilde{B}_t^H \stackrel{(d)}{:=} \lim_{J \rightarrow +\infty} \sum_{j=-\infty}^J 2^{-j/2} \sum_{n=-\infty}^{+\infty} d_j[n] \psi(2^{-j}t - n) \quad (4.2.1)$$

has an averaged spectrum $\mathcal{S}_{\widetilde{B}^H}$ in the sense given by definition ??, which verifies :

$$\forall \omega \in \mathbb{R}, \quad \gamma_1(\psi) \frac{\sigma_H^2}{|\omega|^{2H+1}} \leq \mathcal{S}_{\widetilde{B}^H}(\omega) \leq \gamma_2(\psi) \frac{\sigma_H^2}{|\omega|^{2H+1}} \quad (4.2.2)$$

where $\gamma_1(\psi), \gamma_2(\psi)$ are both constant w.r.t. ω , but dependent on ψ .

Proof. The interested reader is referred to *G. Wornell's* original article [6], where the full proof is presented in detail. $\square$

Remark Because of the assumption made in (4.0.1), (4.2.1) thus provides a *Kosambi-Karhunen-Loève expansion* of the generated process $(\widetilde{B}_t^H)_{t \in \mathbb{R}}$. Moreover, in the proof of Theorem 4.1 presented in [6], it is shown that processes whose AWWSD satisfies :

$$\forall \omega \in \mathbb{R}, \quad \mathcal{S}(\omega) \propto \frac{1}{|\omega|^\gamma}, \quad 1 \leq \gamma \leq 2$$

cannot avoid having an infinite variance, in the sense that the probability of reaching extreme values is non-negligible. This is in contrast to processes with $0 < \gamma < 1$, for which infinite variance can be prevented. Now, since $\gamma = 2H + 1$ falls into the former case, corresponding processes (including fBm) thus necessarily have an average spectrum with infinite variance.

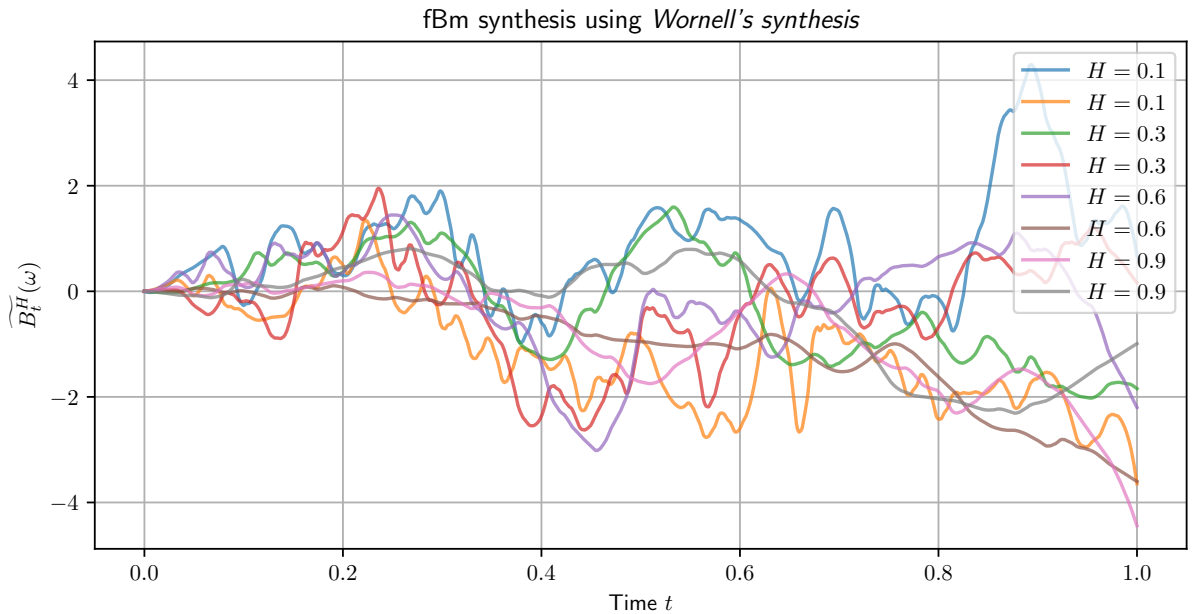


Figure 20: **Multiple realizations of Wornell's orthonormal synthesis of "fBm-like" processes** $(\widetilde{B}_t^H)_{t \in \mathbb{R}}$, constructed via the Karhunen-Loève expansion (4.2.1) using Daubechies-4 wavelets. Results are shown for $H \in \{0.1, 0.3, 0.6, 0.9\}$, with two independent realizations displayed for each Hurst parameter.

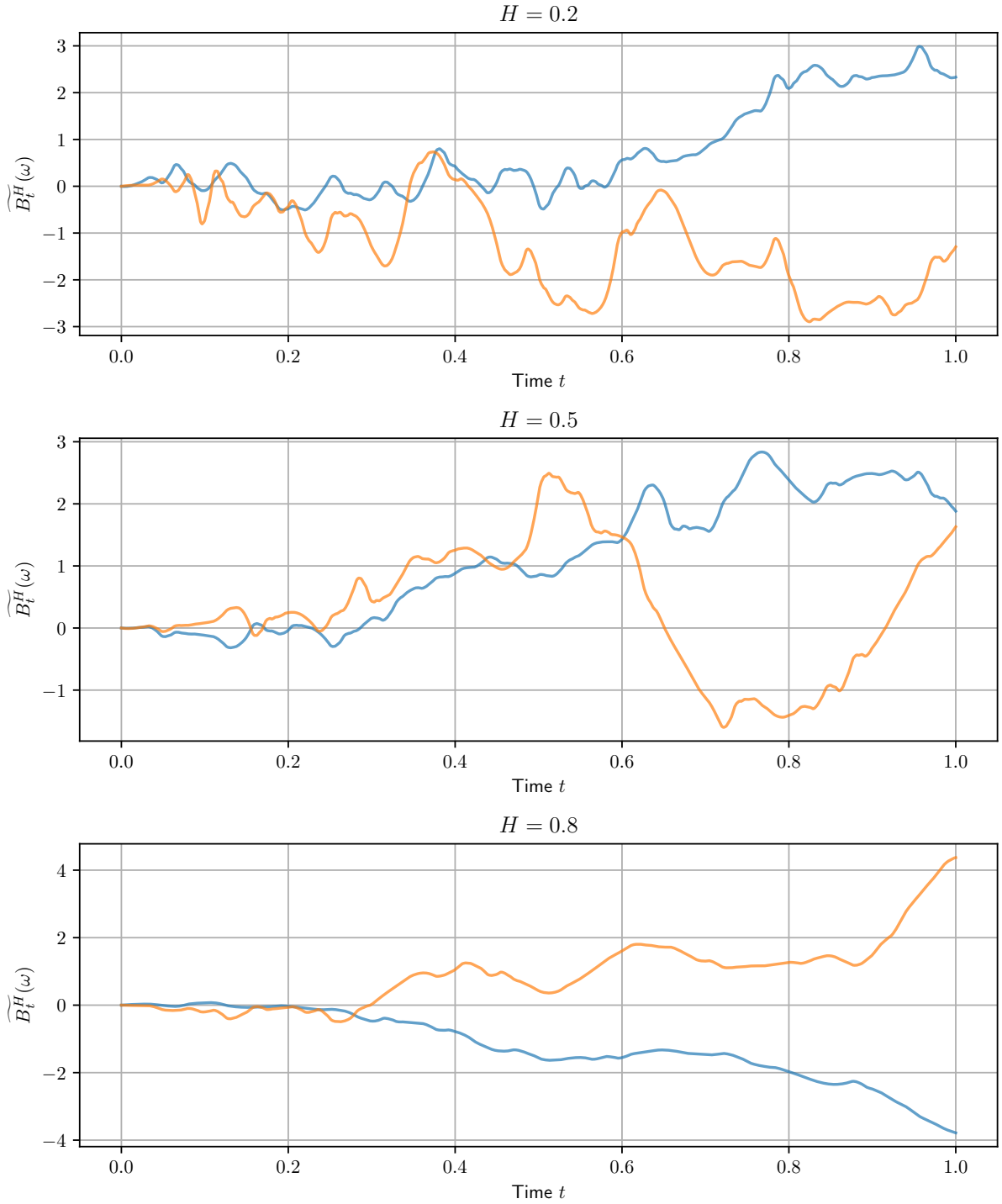


Figure 21: **Individual plots of multiple realizations of Wornell’s orthonormal synthesis of “fBm-like” processes** $(\widetilde{B}_t^H)_{t \in \mathbb{R}}$, constructed via the Karhunen–Loève expansion (4.2.1) using Daubechies-4 wavelets, for $H \in \{0.2, 0.5, 0.8\}$. Two independent realizations are shown for each value of the Hurst parameter.

From what is observed in Figure 21, stochastic processes issued from the *Kosambi–Karhunen–Loève* expansion (4.2.1) behave much more like fractional Brownian motion, in sharp contrast with what was obtained from (4.1.1) using the IDWT. As discussed previously, this similarity should be understood in the averaged-spectrum sense, since these

processes satisfy (4.2.2) and therefore exhibit an effective $1/f$ -type spectrum, comparable to that of fBm.

However, it may be noticeable that the self-similarity of the resulting trajectories, namely their fractal-like structure, appears significantly weakened (compare, for instance, figures 4 and 20). This behavior is naturally explained by the uncorrelation assumption in (4.0.1), since even slight correlations between wavelet detail coefficients are known to contribute substantially to the fractal nature of the sample paths.

As a result, one might argue that the situation should improve when considering mother wavelets ψ with a higher number of vanishing moments R , as discussed in Section 3.3.2. However, it is well known that the synthesis (4.2.1) depends explicitly on the choice of ψ , and in particular on its regularity, which itself is controlled by R . In Section IV.B of [1], *P. Flandrin* indeed argues that the influence of both the regularity of the wavelet shape and the decorrelation assumption should diminish as R increases. Nevertheless, the figures below show that the *G. Wornell synthesis* remains strongly dependent on the shape of ψ . In particular, for extreme values of R (e.g. see figure 25 corresponding to Daubechies–25 wavelets), the resulting processes differ significantly from fBms.

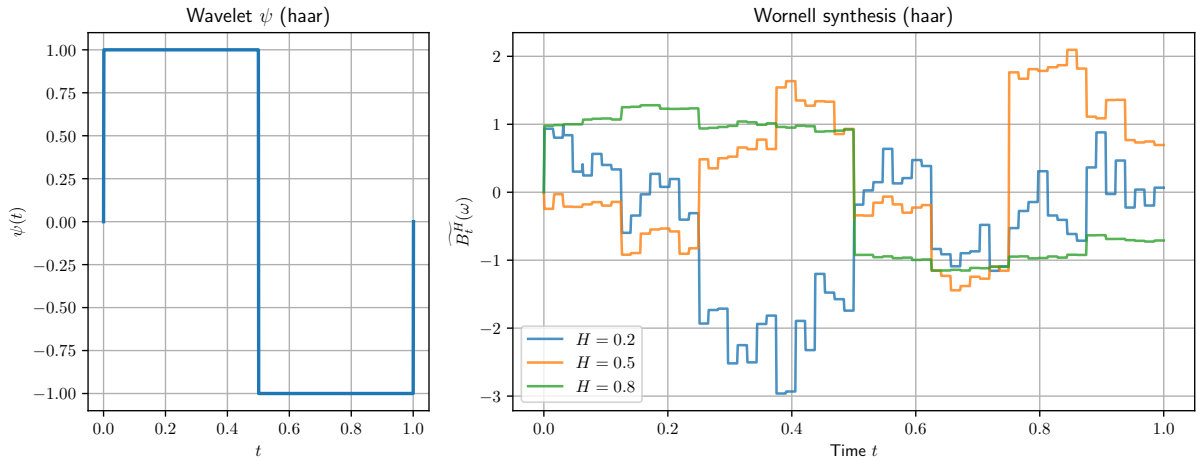


Figure 22: **Wornell synthesis using the Haar wavelet ($R = 1$).** Left : Haar mother wavelet ψ . Right : Individual realizations of the Wornell orthonormal synthesis of “fBm-like” processes $(\widetilde{B}_t^H)_{t \in \mathbb{R}}$ for $H \in \{0.2, 0.5, 0.8\}$. The strong irregularity of the wavelet results in trajectories that significantly depart from the expected behavior of fBm.

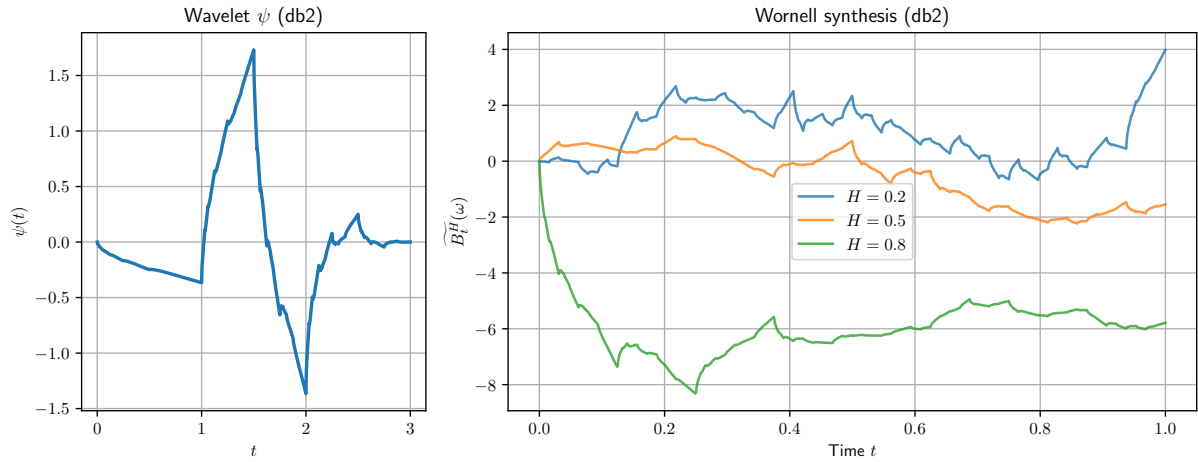


Figure 23: **Wornell synthesis using the Daubechies-2 wavelet ($R = 2$)**. Left : Daubechies-2 mother wavelet ψ . Right : Individual realizations of the Wornell orthonormal synthesis of “fBm-like” processes $(\tilde{B}_t^H)_{t \in \mathbb{R}}$ for $H \in \{0.2, 0.5, 0.8\}$. Increasing the number of vanishing moments improves regularity but the synthesized processes still exhibit noticeable deviations from true fractional Brownian motion.

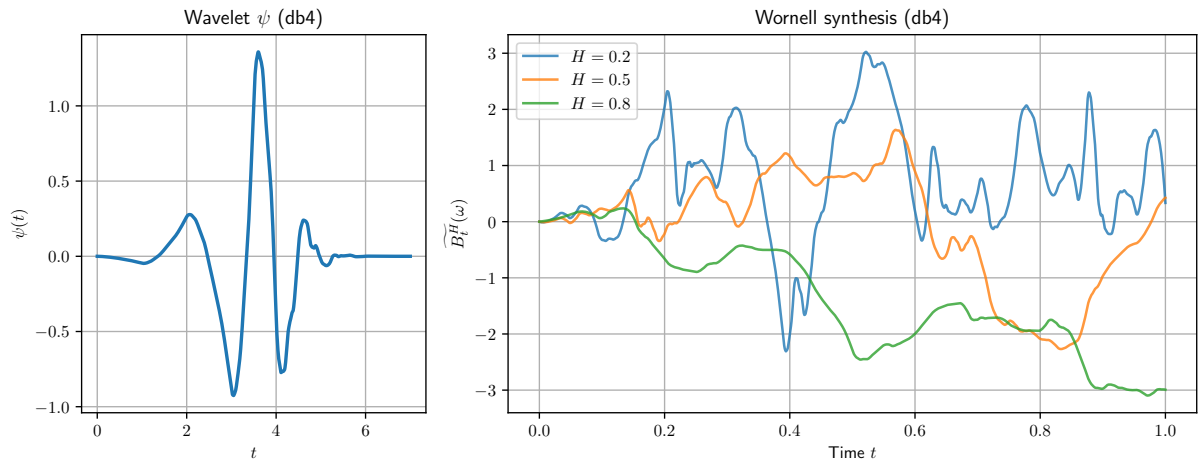


Figure 24: **Wornell synthesis using the Daubechies-4 wavelet ($R = 4$)**. Left : Daubechies-4 mother wavelet ψ . Right : Individual realizations of the Wornell orthonormal synthesis of “fBm-like” processes $(\tilde{B}_t^H)_{t \in \mathbb{R}}$ for $H \in \{0.2, 0.5, 0.8\}$. The increased regularity of ψ yields trajectories that more closely resemble fBm, though a residual dependence on the wavelet shape remains visible.

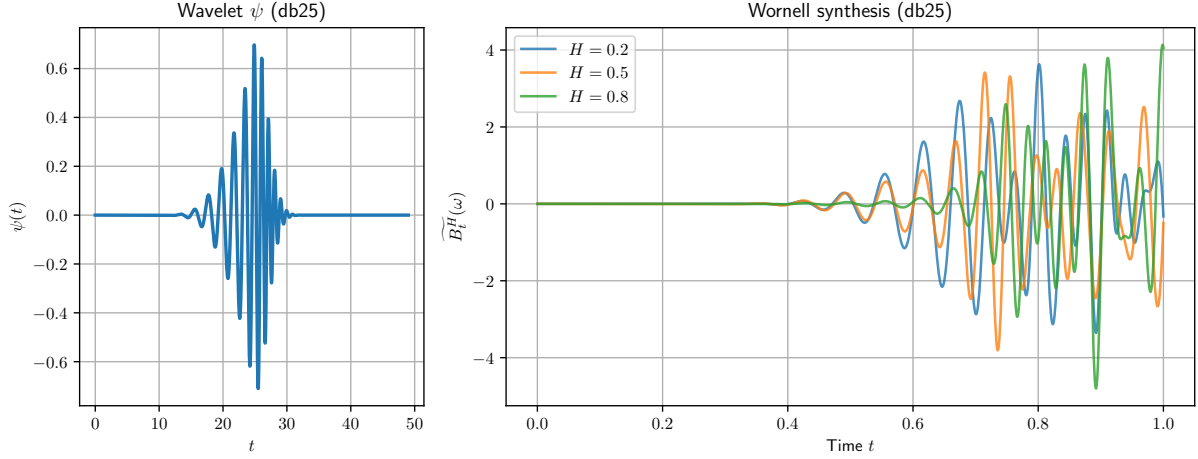


Figure 25: **Wornell synthesis using the Daubechies-25 wavelet ($R = 25$)**. Left : Daubechies-25 mother wavelet ψ . Right : Individual realizations of the Wornell orthonormal synthesis of “fBm-like” processes $(\tilde{B}_t^H)_{t \in \mathbb{R}}$ for $H \in \{0.2, 0.5, 0.8\}$. Despite the high number of vanishing moments and increased smoothness of ψ , the synthesized processes exhibit marked deviations from fBm, highlighting the strong dependence of the Wornell synthesis on the wavelet shape at extreme regularity levels.

4.3 Random Midpoint Displacement Method (RMD)

A widely used method to synthesize discrete trajectories resembling fractional Brownian motions is the *Random Midpoint Displacement (RMD)* algorithm. The idea is to generate a sequence of points on the interval $[0, 1]$ recursively, using dyadic subdivisions and Gaussian displacements whose variance depends on the interval length, in order to reproduce the scaling properties of a fractional Brownian motion with Hurst parameter $H \in (0, 1)$.

The algorithm proceeds as follows :

1. **Initialization** : Set the first point $B[0] = 0$ and the last point $B[1] \sim \mathcal{N}(0, \sigma_H^2)$.
2. **Dyadic subdivision** : For each interval $[t_i, t_{i+1}]$, compute the midpoint :

$$t_m := \frac{t_i + t_{i+1}}{2}$$

and assign :

$$B[t_m] := \frac{B[t_i] + B[t_{i+1}]}{2} + \varepsilon, \quad \varepsilon \sim \mathcal{N}\left(0, \frac{|t_{i+1} - t_i|^{2H}}{2}\right) \quad (4.3.1)$$

3. **Recursion** : Repeat the previous step on each newly formed subinterval until the desired resolution, determined by the number of dyadic steps J , is reached. After J subdivisions, the trajectory consists of $2^J + 1$ points.

4. **Normalization (optional)** : Center and normalize the trajectory :

$$B \leftarrow \frac{B - B[0]}{\text{std}(B)} = \frac{B}{\text{std}(B)} \quad (4.3.2)$$

This construction ensures that the generated trajectory $(B[t])_{t \in [0,1]}$ exhibits the approximate scaling property characteristic of a fractional Brownian motion :

$$\text{Var}(B[t + \tau] - B[t]) \sim \sigma_H^2 |\tau|^{2H}. \quad (4.3.3)$$

However, as noted by *P. Flandrin* in [1], it is of much importance to state that the RMD method fails to generate processes with stationary increments when $H \neq 1/2$.

Algorithm 1 Random Midpoint Displacement for Fractional Brownian Motion

Require: Hurst parameter $H \in (0, 1)$, number of dyadic steps J , and desired endpoint variance σ_H^2

Ensure: Trajectory $B_t^H(\omega)$ with $2^J + 1$ points, $t \in [0, 1]$

- 1: Initialize array B of size $N = 2^J + 1$
 - 2: Set endpoints : $B[0] \leftarrow 0, B[N - 1] \sim \mathcal{N}(0, \sigma_H^2)$
 - 3: Set initial step size : $step \leftarrow N - 1$
 - 4: Set initial scale factor : $scale \leftarrow 1$
 - 5: **while** $step > 1$ **do**
 - 6: $half \leftarrow step/2$
 - 7: **for** $k = 0$ **to** $N - 1$ **step** $step$ **do**
 - 8: $mid \leftarrow k + half$
 - 9: Compute midpoint mean : $mean \leftarrow (B[k] + B[k + step]) / 2$
 - 10: Compute standard deviation : $\sigma \leftarrow (\sigma_H \cdot scale^H) / \sqrt{2}$
 - 11: Draw Gaussian displacement : $\varepsilon \sim \mathcal{N}(0, \sigma^2)$
 - 12: Set midpoint : $B[mid] \leftarrow mean + \varepsilon$
 - 13: **end for**
 - 14: Reduce step size : $step \leftarrow step/2$
 - 15: Reduce scale : $scale \leftarrow scale/2$
 - 16: **end while**
 - 17: Normalize trajectory : $B \leftarrow B/\text{std}(B)$
 - 18: **return** B
-

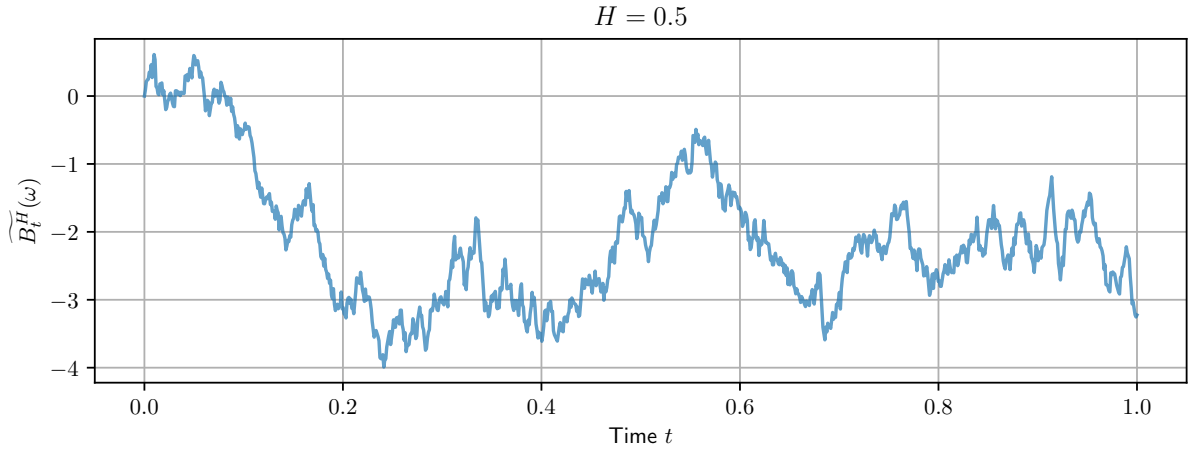


Figure 26: **Single realization of a fBm synthesized using the Random Midpoint Displacement (RMD) method.** The trajectory corresponds to $H = 0.5$, with $2^{10} + 1$ points and endpoint variance σ_H^2 . Moreover, the process happens to have stationary increments as $H = 1/2$.

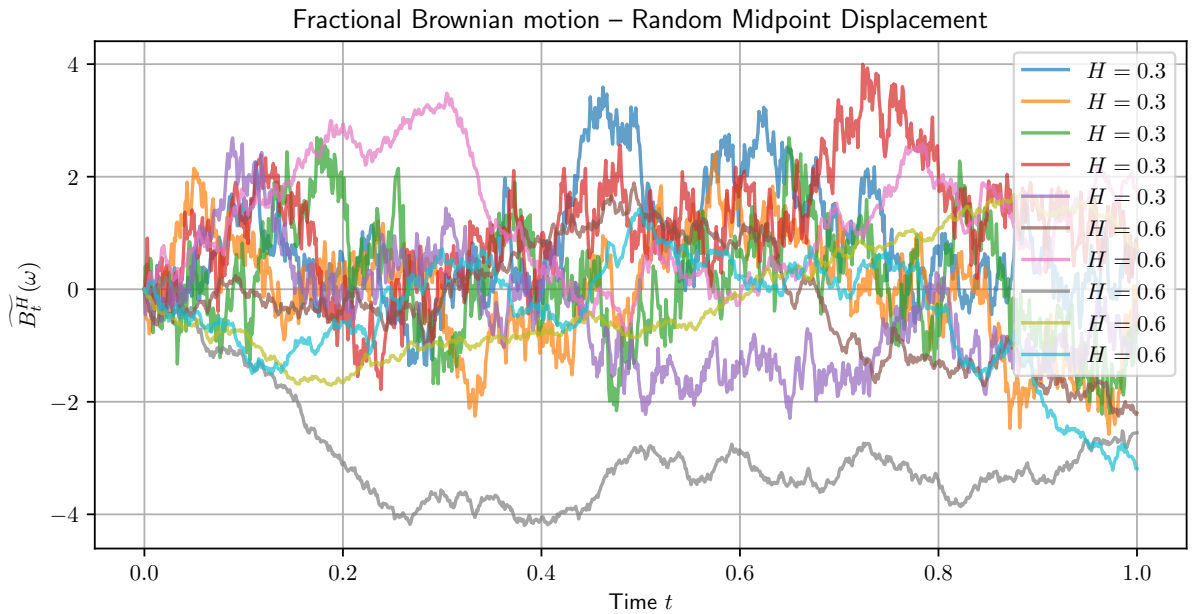


Figure 27: **Multiple realizations of fBms on a single plot, synthesized using the RMD method.** Trajectories correspond to Hurst parameters $H \in \{0.3, 0.6\}$, each with $2^{10} + 1$ points and endpoint variance σ_H^2 .

What we observe is that running algorithm 1 yields very satisfactory results in terms of reproducing the overall shape and visual behaviour of a fractional Brownian motion (see, for instance, Figures 4 and 27). The expected self-similar regularity effect is also recovered when varying the Hurst parameter H , as illustrated in Figure 28. However, as discussed previously, the processes generated by the Random Midpoint Displacement method do not possess stationary increments unless $H = 1/2$, which constitutes a significant departure from the defining statistical properties of fBms.



Figure 28: **Multiple realizations of fBms synthesized using the RMD method.** Each subplot shows one realization for Hurst parameters $H \in \{0.2, 0.5, 0.8\}$, with $2^{10} + 1$ points and endpoint variance σ_H^2 .

Although the Random Midpoint Displacement (RMD) method does not directly make use of wavelet coefficients, *P. Flandrin* [1] claims that it can be interpreted within a multiresolution framework.

Let $a_j[n]$ denote the coarse-scale approximation at scale j , and $d_j[n]$ the detail

coefficients. The recursive synthesis from scale $j + 1$ to j can then be written as:

$$a_j[n] = a_{j+1} \left[\frac{n}{2} \right] + \frac{1}{2} \left(a_{j+1} \left[\frac{n-1}{2} \right] + a_{j+1} \left[\frac{n+1}{2} \right] \right) + d_{j+1} \left[\frac{n+1}{2} \right] \quad (4.3.4)$$

This shows how the coarse-scale coefficient $a_j[n]$ is obtained from neighboring fine-scale values a_{j+1} and the random detail d_{j+1} . In fact, (4.3.4) exactly reflects what is done in the RMD algorithm 1, where n is the midpoint index.

Here, the averaging corresponds to a low-pass filter :

$$h[0] = 1, \quad h[-1] = h[1] = \frac{1}{2}$$

and the random increment corresponds to a high-pass filter :

$$g[1] = 1$$

Recursively applying this across scales reproduces the hierarchical structure of RMD. However, this system does not form an orthonormal basis, since the filters overlap in a way that prevents orthogonality. Nonetheless, this interpretation highlights the equivalence between classical RMD and a dyadic, wavelet-like synthesis of a fractal process.

5 Conclusion

In this study, we have shown that wavelet analysis provides a powerful and coherent framework for understanding the fundamental properties of fractional Brownian motion. By exploiting the intrinsic multiscale structure of wavelets, key characteristics of fBm—such as self-similarity, long-range dependence, and scale-dependent variance—can be directly linked to the statistical behavior of wavelet coefficients. In particular, the stationarity of detail coefficients across time contrasts with the non-stationary nature of the process itself and plays a central role in the practical estimation of the Hurst exponent.

The log-scale estimation method based on the discrete wavelet transform was shown to be both effective and robust over a wide range of Hurst parameters, while the continuous wavelet approach suffers from bias due to the mixing of low- and high-frequency components. Our analysis of correlation decay further highlighted the importance of the chosen wavelet basis, demonstrating how vanishing moments strongly influence the asymptotic decorrelation of coefficients and, consequently, the efficiency of statistical inference.

On the synthesis side, wavelet-based generation methods offer an intuitive and computationally efficient way to construct fBm-like processes. While inverse wavelet reconstruction and Wornell's synthesis capture essential scaling properties, they may still exhibit deviations from true fractional Brownian motion, especially for extreme regularity levels. The Random Midpoint Displacement method, although visually convincing and interpretable within a multiresolution framework, fails to preserve stationary increments except in the Brownian case. Overall, these results emphasize both the theoretical insight and the practical relevance of wavelet methods for analyzing and synthesizing fractional Brownian motion, while also underlining the limitations that must be considered when approximating such stochastic processes.

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